

ML Final project

This section is focusing on project of Should we upgrade our claim severity model

Step 1: Load the personal auto claim and commercial property claim data sets

```
library(tidyverse)
```

Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

Warning: package 'stringr' was built under R version 4.5.2

Warning: package 'forcats' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr 1.1.4 ✓ readr 2.1.6

✓ forcats 1.0.1 ✓ stringr 1.6.0

✓ ggplot2 4.0.1 ✓ tibble 3.3.1

✓ lubridate 1.9.4 ✓ tidyr 1.3.2

✓ purrr 1.2.1

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

ℹ Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become errors

```
p_auto <- read.csv('personal_auto_cv.csv')
c_property <- read.csv('commercial_property_cv.csv')
```

Step 2: Inspect the personal claim dataset

```
head(p_auto)
```

	claim_id	true_cost	pred_A	pred_B	fold
1	1	1468.94	0.00	0.00	3
2	2	3995.84	0.00	0.00	8
3	3	2938.10	8032.39	3332.58	7
4	4	4279.00	4135.48	5080.83	2
5	5	2614.50	0.00	3098.96	6
6	6	3340.58	4183.44	4118.76	1

```
glimpse(p_auto)
```

Rows: 12,000

Columns: 5

```
$ claim_id <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1...
$ true_cost <dbl> 1468.94, 3995.84, 2938.10, 4279.00, 2614.50, 3340.58, 1393.3...
$ pred_A <dbl> 0.00, 0.00, 8032.39, 4135.48, 0.00, 4183.44, 8099.95, 3095.4...
$ pred_B <dbl> 0.00, 0.00, 3332.58, 5080.83, 3098.96, 4118.76, 4162.57, 137...
$ fold <int> 3, 8, 7, 2, 6, 1, 1, 6, 7, 5, 4, 6, 3, 7, 4, 6, 3, 10, 9, 5,...
```

Step 2B: observed some prediction values are 0, so decided to check how frequent this is

```
p_auto %>%
  summarise(
    prop_zero_A = mean(pred_A == 0),
    prop_zero_B = mean(pred_B == 0)
  )
```

	prop_zero_A	prop_zero_B
1	0.18275	0.1279167

Step 3: Inspect the commercial property claim dataset

```
head(c_property)
```

	claim_id	true_cost	pred_A	pred_B	fold
1	1	6348.04	12960.79	12013.24	9
2	2	3408.67	0.00	0.00	9
3	3	63619.69	27794.20	26772.71	8
4	4	10374.75	3000.69	29628.03	8
5	5	8002.53	20227.52	4464.30	10
6	6	31445.62	12774.43	12976.06	4

```
glimpse(c_property)
```

```
Rows: 4,500
```

```
Columns: 5
```

```
$ claim_id <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 1...
$ true_cost <dbl> 6348.04, 3408.67, 63619.69, 10374.75, 8002.53, 31445.62, 190...
$ pred_A <dbl> 12960.79, 0.00, 27794.20, 3000.69, 20227.52, 12774.43, 29389...
$ pred_B <dbl> 12013.24, 0.00, 26772.71, 29628.03, 4464.30, 12976.06, 18987...
$ fold <int> 9, 9, 8, 8, 10, 4, 5, 7, 3, 3, 6, 1, 3, 5, 4, 4, 4, 8, 2, 7,...
```

```
c_property %>%
  summarise(
    prop_zero_A = mean(pred_A == 0),
    prop_zero_B = mean(pred_B == 0)
  )
```

```
prop_zero_A prop_zero_B
1 0.2895556 0.2691111
```

Observation: Both models produce a noticeable proportion of zero predictions, particularly in the commercial property dataset. However, since evaluation is based on MAE, the impact of these predictions is appropriately reflected in the performance metric.

Step 4: Quick diagnostic to confirm if many actual claims are small -> then zeros makes sense, if not, -> model might be underpredicting

```
mean(p_auto$true_cost < 100)
```

```
[1] 0
```

```
mean(c_property$true_cost < 100)
```

```
[1] 0
```

Observation note: A noticeable proportion of predictions from both models are zero, despite no observed claims being near zero. This suggests potential underprediction for a subset of cases. However,

since MAE directly penalizes these errors, their impact is appropriately reflected in the model comparison.

Step 5: Quick deeper diagnostic check to confirm that predictions at zero are not realistic predictions

```
mean(p_auto$pred_A == 0 & p_auto$true_cost > 1000)
```

```
[1] 0.1649167
```

```
mean(p_auto$pred_B == 0 & p_auto$true_cost > 1000)
```

```
[1] 0.1126667
```

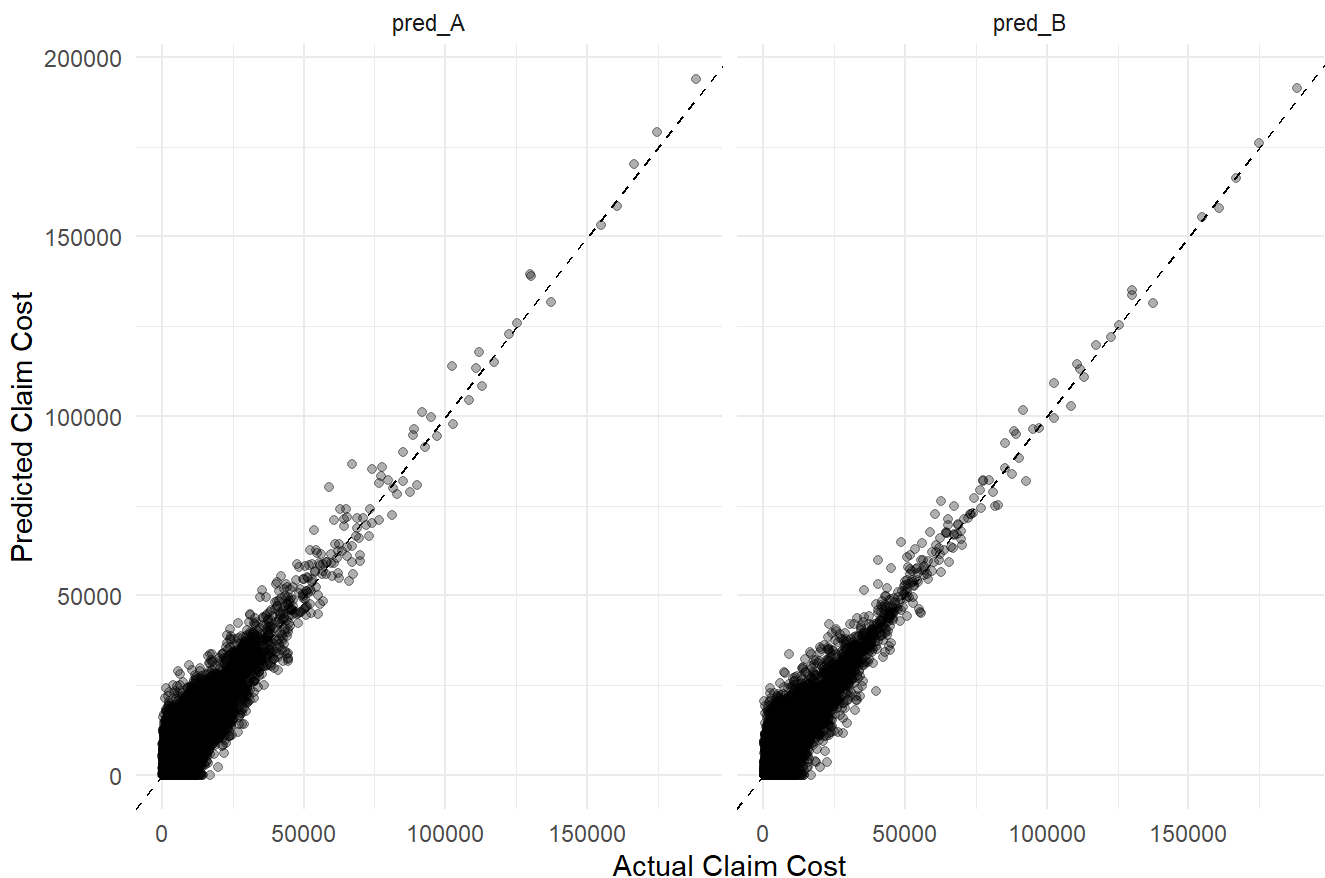
Observation: Both models have an underestimation problem because they are missing a chunk of meaningful claims. Model B reduces the frequency of severe underprediction cases compared to Model A, indicating improved handling of higher-cost claims.

Step 6: Create a main visual (scatter plot) of actual vs predicted for model A and model B using personal auto claim dataset

```
plot_data <- p_auto %>%
  select(true_cost, pred_A, pred_B) %>%
  pivot_longer(
    cols = c(pred_A, pred_B),
    names_to = "model",
    values_to = "predicted_cost"
  )

ggplot(plot_data, aes(x = true_cost, y = predicted_cost)) +
  geom_point(alpha = 0.3) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  facet_wrap(~ model) +
  labs(
    title = "Actual vs Predicted Claim Cost by Model",
    x = "Actual Claim Cost",
    y = "Predicted Claim Cost"
  ) +
  theme_minimal()
```

Actual vs Predicted Claim Cost by Model



Observation note: From the scatter plot above, distance to the 45-degree line: model B looks slightly tighter (better predictions) around the line, and model A looks a bit more spread (larger errors). Underprediction pattern: Model B appears slightly less below the line (less severe underprediction). Spread of points: Model A is a bit more dispersed, and model B is more concentrated, less spread-> lower error variance. Model B appears to produce predictions that are more closely aligned with actual values, suggesting improved performance, though this needs to be confirmed using MAE.

Step 7: Create supporting visual (density plot of residuals)

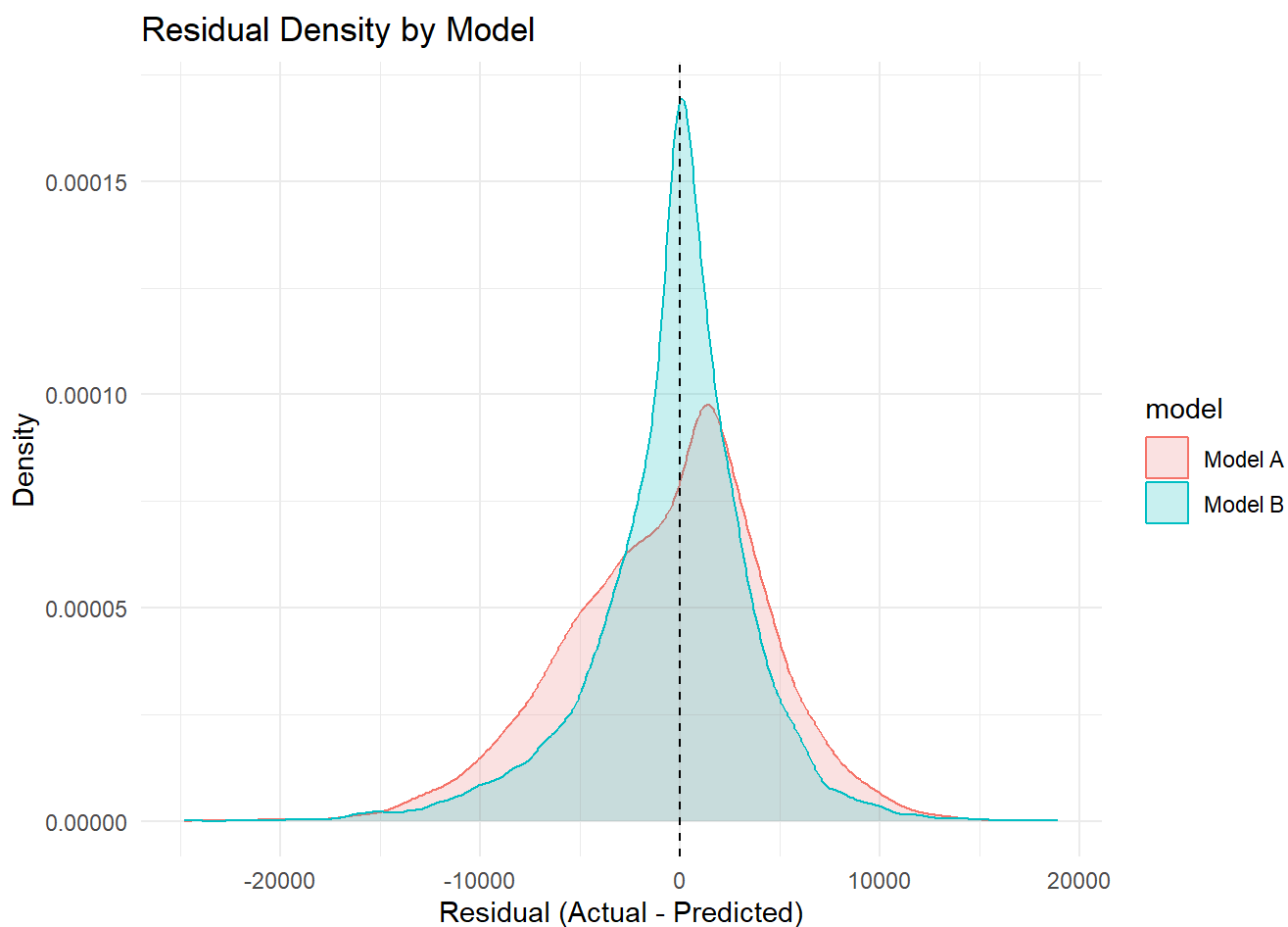
```
p_auto <- p_auto %>%
  mutate(
    resid_A = true_cost - pred_A,
    resid_B = true_cost - pred_B
  )
resid_data <- p_auto %>%
```

```

select(resid_A, resid_B) %>%
pivot_longer(
  cols = everything(),
  names_to = "model",
  values_to = "residual"
) %>%
mutate(
  model = recode(model,
    resid_A = "Model A",
    resid_B = "Model B")
)

ggplot(resid_data, aes(x = residual, color = model, fill = model)) +
  geom_density(alpha = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Residual Density by Model",
    x = "Residual (Actual - Predicted)",
    y = "Density"
  ) +
  theme_minimal()

```



Observation from the density plot by model: Model B is more precise because its errors are more concentrated near zero. Model A has fatter tails, especially on the right side (underprediction), which means that model A makes more large errors, and more extreme underprediction cases. Overall, both models are centered around 0 -> not heavily biased, but model B has tighter distribution, fewer extreme errors, more consistent predictions. Model A on the other hand, wider spread, more larger errors, and more variability.

step 8: Compute MAE per fold

```
library(dplyr)

mae_by_fold <- p_auto %>%
  group_by(fold) %>%
  summarise(
    MAE_A = mean(abs(true_cost - pred_A)),
    MAE_B = mean(abs(true_cost - pred_B))
  )
mae_by_fold
```

```
# A tibble: 10 × 3
  fold MAE_A MAE_B
<int> <dbl> <dbl>
1     1 3971. 2101.
2     2 4043. 2662.
3     3 3990. 1691.
4     4 3836. 2982.
5     5 3752. 4336.
6     6 3948. 1297.
7     7 3901.  394.
8     8 3980. 4321.
9     9 3772. 4762.
10    10 3905. 3701.
```

Step 9: Compute improvement by fold

```
mae_by_fold <- mae_by_fold %>%
  mutate(
    improvement = MAE_B - MAE_A
  )
mae_by_fold
```

```
# A tibble: 10 × 4
  fold MAE_A MAE_B improvement
  <int> <dbl> <dbl>         <dbl>
1     1 3971. 2101.         -1870.
2     2 4043. 2662.         -1381.
3     3 3990. 1691.         -2299.
4     4 3836. 2982.          -854.
5     5 3752. 4336.           583.
6     6 3948. 1297.        -2651.
7     7 3901.  394.        -3507.
8     8 3980. 4321.           341.
9     9 3772. 4762.           990.
10    10 3905. 3701.         -204.
```

observation from the table above, model B is better in 7 out of 10 folds, model A is better in 3 folds. This indicates that the improvement is not perfect, but is mostly consistent.

Step 10: Summarize overall performance

```
summary_stats <- mae_by_fold %>%
  summarise(
    mean_MAE_A = mean(MAE_A),
    mean_MAE_B = mean(MAE_B),
    mean_improvement = mean(improvement),
    sd_improvement = sd(improvement)
  )

summary_stats
```

```
# A tibble: 1 × 4
  mean_MAE_A mean_MAE_B mean_improvement sd_improvement
  <dbl>         <dbl>         <dbl>         <dbl>
1    3910.         2825.         -1085.         1507.
```

Observation from the mean MAE table, the mean improvement is 1085.36, which means that model B reduces error by about \$1,085 per claim. there is noticeable variability in improvement across folds, as reflected by a standard deviation larger than the mean. However, the improvement remains consistently favorable overall.

Step 11: Confidence Interval

```
n <- nrow(mae_by_fold)

se <- summary_stats$sd_improvement / sqrt(n)

lower <- summary_stats$mean_improvement - 2 * se
upper <- summary_stats$mean_improvement + 2 * se

c(lower, upper)
```

```
[1] -2038.5874 -132.1269
```

Observation from the confidence interval, the entire interval is below 0, which means even in the worst-case estimate, model B is still better. Therefore, improvement is statistically reliable. Summary of the overall evaluation: Model B reduces mean absolute error by approximately \$1,085 per claim relative to Model A. This improvement is observed in the majority of cross-validation folds and remains statistically reliable, with the confidence interval entirely below zero. These results indicate that Model B provides a meaningful and consistent improvement in predictive performance. In addition, this improvement is consistent with earlier diagnostic findings, where Model B exhibited tighter error distributions and reduced variability.

Step 12: Creating confidence interval plot using the personal auto claim data

```
library(ggplot2)

summary_df <- data.frame(
  x = 5.5,
  mean = summary_stats$mean_improvement,
  lower = lower,
  upper = upper
)

ggplot(mae_by_fold, aes(x = factor(fold), y = improvement)) +
  geom_point(size = 3, color = "#2C7BB6") +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
```

```
# Mean point
geom_point(
  data = summary_df,
  aes(x = x, y = mean),
  inherit.aes = FALSE,
  size = 5,
  color = "black"
) +

# Confidence interval
geom_errorbar(
  data = summary_df,
  aes(x = x, ymin = lower, ymax = upper),
  inherit.aes = FALSE,
  width = 0.2,
  color = "black",
  size = 1.2
) +

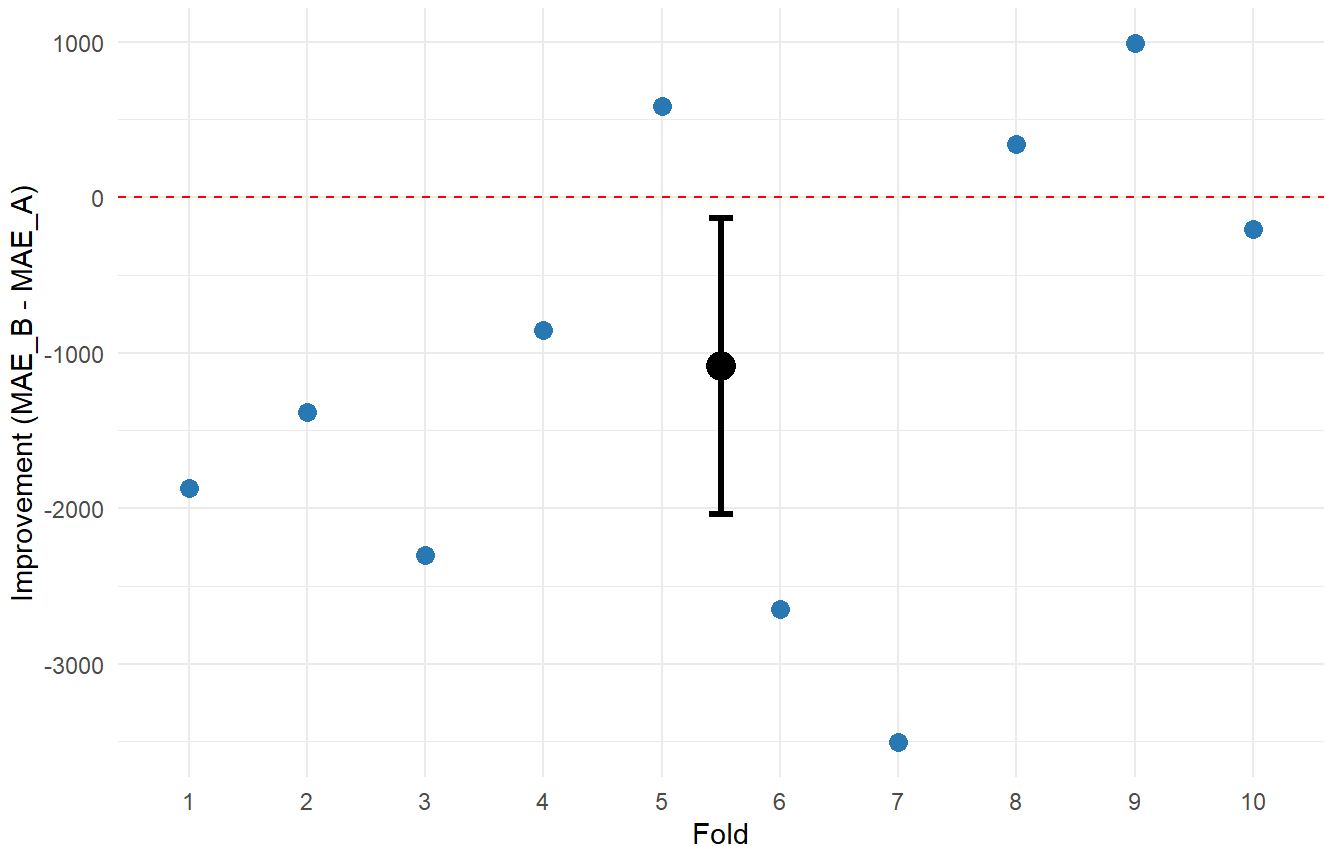
labs(
  title = "Model Improvement with Confidence Interval",
  subtitle = "All estimates fall below zero, indicating consistent improvement",
  x = "Fold",
  y = "Improvement (MAE_B - MAE_A)"
) +
theme_minimal()
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

• Please use `linewidth` instead.

Model Improvement with Confidence Interval

All estimates fall below zero, indicating consistent improvement

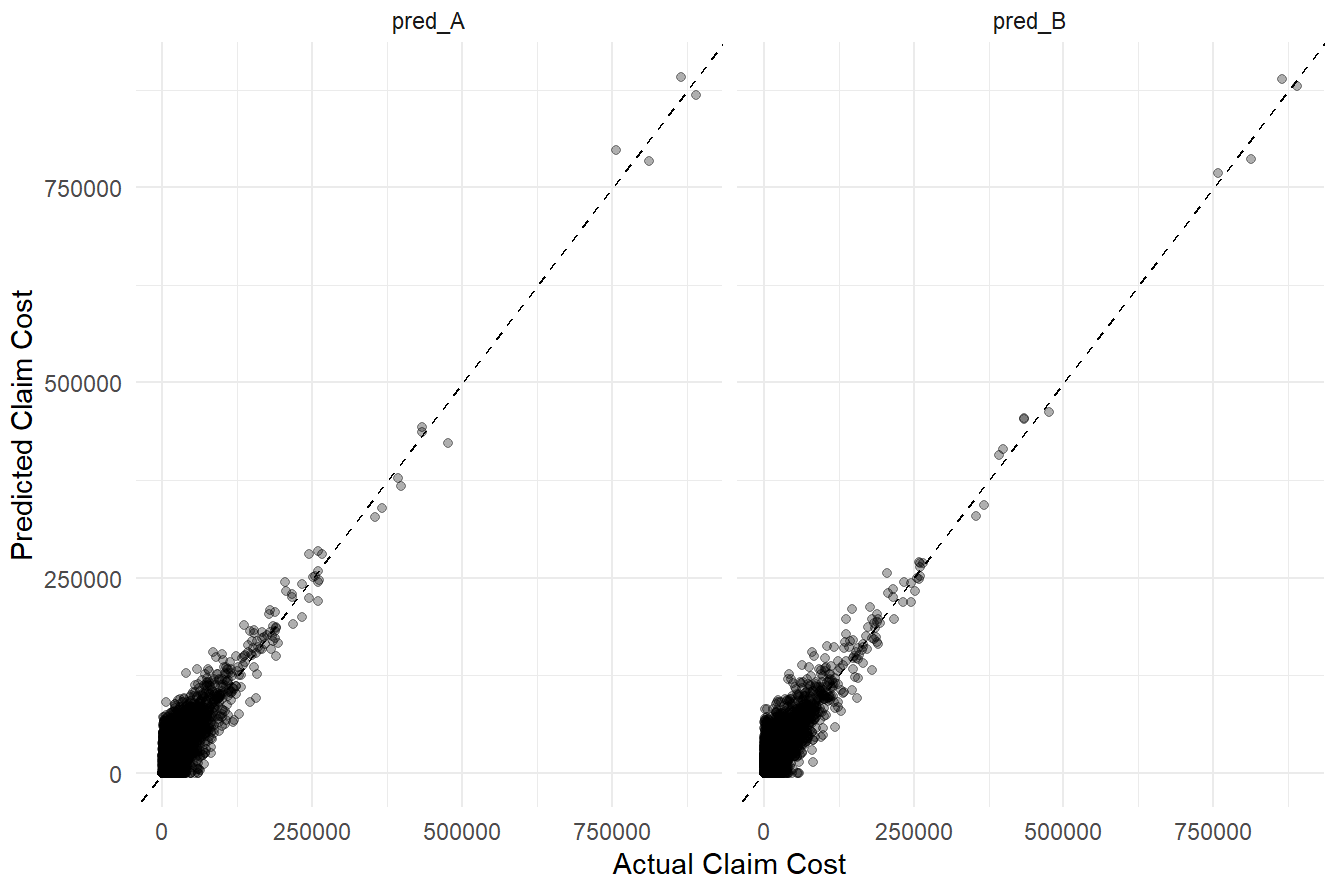


Step 13: Create scatter plot using commercial property claim dataset to gain some diagnostic insights

```
plot_data <- c_property %>%
  select(true_cost, pred_A, pred_B) %>%
  pivot_longer(
    cols = c(pred_A, pred_B),
    names_to = "model",
    values_to = "predicted_cost"
  )

ggplot(plot_data, aes(x = true_cost, y = predicted_cost)) +
  geom_point(alpha = 0.3) +
  geom_abline(intercept = 0, slope = 1, linetype = "dashed") +
  facet_wrap(~ model) +
  labs(
    title = "Actual vs Predicted Claim Cost by Model",
    x = "Actual Claim Cost",
    y = "Predicted Claim Cost"
  ) +
  theme_minimal()
```

Actual vs Predicted Claim Cost by Model



Observation: Model B shows tighter alignment with actual values compared to Model A. However, due to the larger scale and heavy-tailed nature of the commercial property data, both models exhibit substantial error, especially for high-cost claims.

Step 14: Create supporting visual (density plot of residuals)

```
c_property <- c_property %>%
  mutate(
    resid_A = true_cost - pred_A,
    resid_B = true_cost - pred_B
  )
resid_data <- c_property %>%
  select(resid_A, resid_B) %>%
  pivot_longer(
    cols = everything(),
    names_to = "model",
    values_to = "residual"
  ) %>%
  mutate(
```

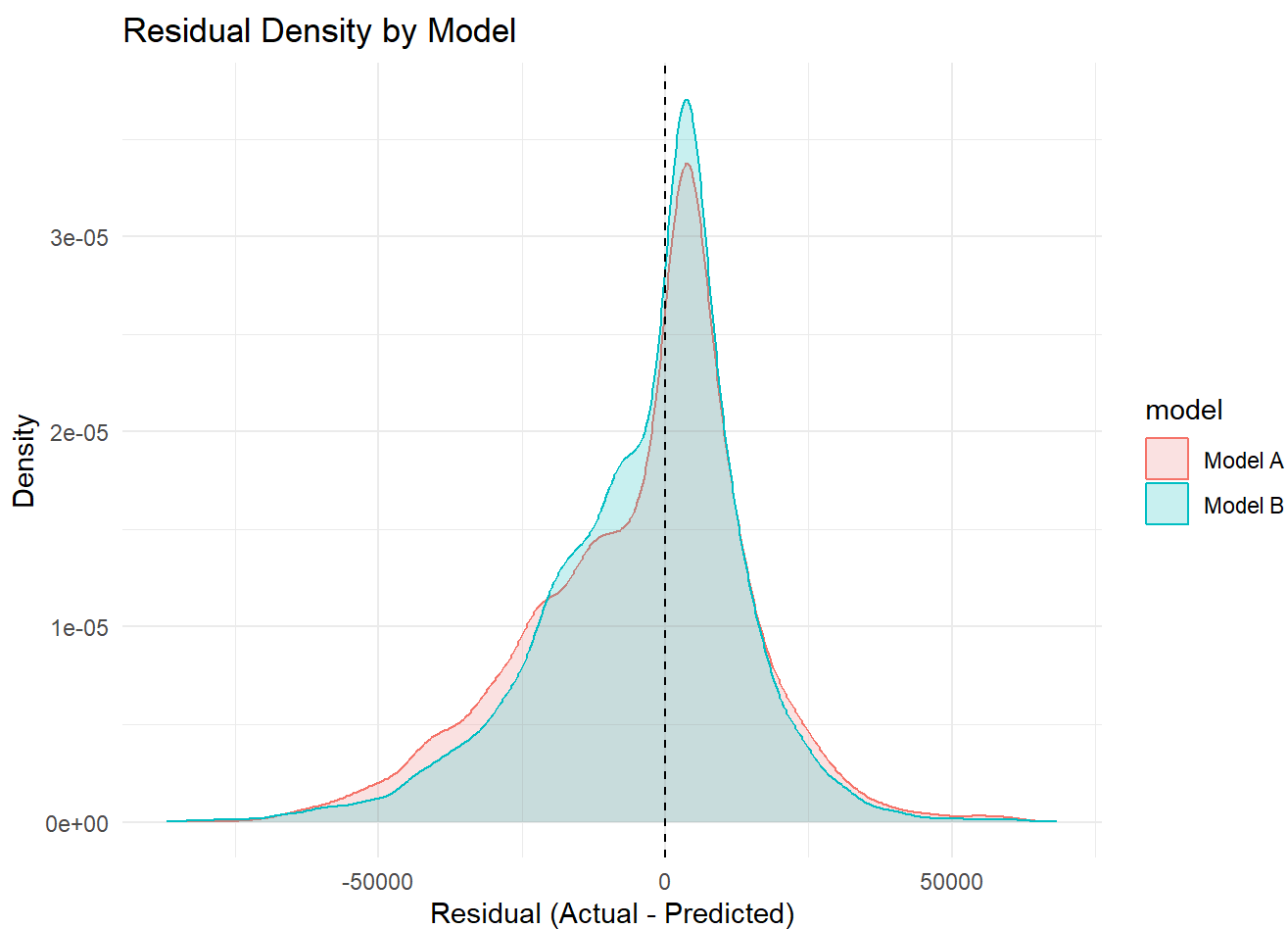
```

model = recode(model,
  resid_A = "Model A",
  resid_B = "Model B")

)

ggplot(resid_data, aes(x = residual, color = model, fill = model)) +
  geom_density(alpha = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  labs(
    title = "Residual Density by Model",
    x = "Residual (Actual - Predicted)",
    y = "Density"
  ) +
  theme_minimal()

```



Step 15: Compute MAE per fold

```

library(dplyr)

mae_by_fold <- c_property %>%
  group_by(fold) %>%
  summarise(

```

```

    MAE_A = mean(abs(true_cost - pred_A)),
    MAE_B = mean(abs(true_cost - pred_B))
  )
mae_by_fold

```

```

# A tibble: 10 × 3
  fold  MAE_A  MAE_B
<int> <dbl> <dbl>
1     1 15217. 10798.
2     2 15512. 17376.
3     3 13968. 16074.
4     4 15364. 12891.
5     5 14842. 10558.
6     6 14404. 15159.
7     7 14811. 15309.
8     8 14193. 13787.
9     9 13590.  9951.
10    10 13752.  7760.

```

Step 16: Compute model improvement by fold

```

mae_by_fold <- mae_by_fold %>%
  mutate(
    improvement = MAE_B - MAE_A
  )
mae_by_fold

```

```

# A tibble: 10 × 4
  fold  MAE_A  MAE_B improvement
<int> <dbl> <dbl>      <dbl>
1     1 15217. 10798.    -4419.
2     2 15512. 17376.     1864.
3     3 13968. 16074.      2106.
4     4 15364. 12891.    -2473.
5     5 14842. 10558.    -4283.
6     6 14404. 15159.       755.
7     7 14811. 15309.       498.
8     8 14193. 13787.     -407.
9     9 13590.  9951.    -3639.
10    10 13752.  7760.    -5992.

```

observation: Model B outperforms Model A in most folds, and when it does, the improvement tends to be substantial. However, it underperforms in a non-trivial number of folds, indicating variability in performance. This dataset exhibits more variability in model

performance, with Model B not consistently outperforming Model A across all folds.

Step 17: Summarize overall performance

```
summary_stats <- mae_by_fold %>%
  summarise(
    mean_MAE_A = mean(MAE_A),
    mean_MAE_B = mean(MAE_B),
    mean_improvement = mean(improvement),
    sd_improvement = sd(improvement)
  )

summary_stats
```

```
# A tibble: 1 × 4
  mean_MAE_A mean_MAE_B mean_improvement sd_improvement
    <dbl>      <dbl>          <dbl>          <dbl>
1    14565.    12966.          -1599.          2915.
```

Step 18: Confidence interval

```
n <- nrow(mae_by_fold)

se <- summary_stats$sd_improvement / sqrt(n)

lower <- summary_stats$mean_improvement - 2 * se
upper <- summary_stats$mean_improvement + 2 * se

c(lower, upper)
```

```
[1] -3442.3687 244.3816
```

Observation: While Model B shows a larger average reduction in MAE, the confidence interval includes zero, indicating that the improvement is not statistically reliable. which indicates that Model B may not outperform Model A in all cases.

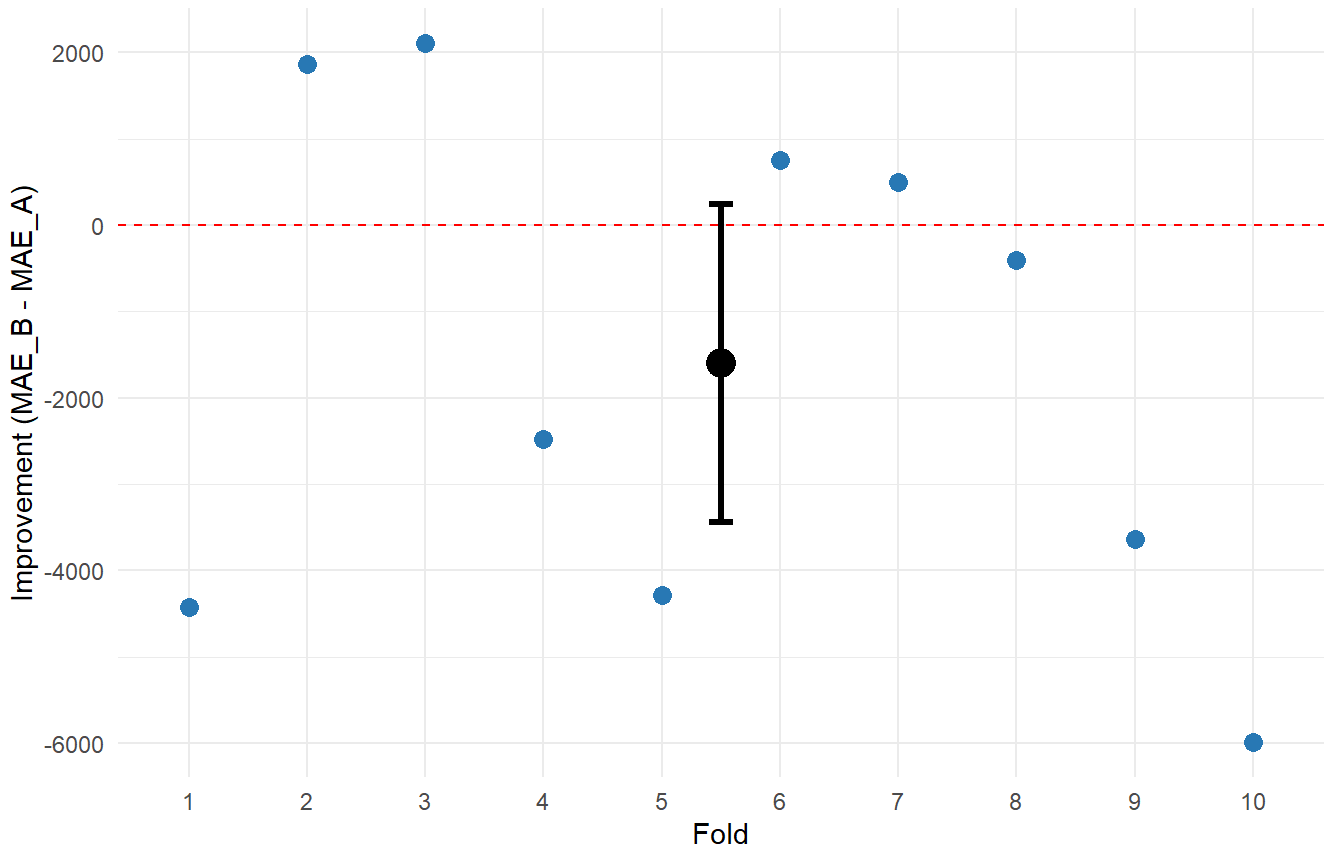
Step 19: Confidence interval plot

```
library(ggplot2)
```

```
summary_df <- data.frame(  
  x = 5.5,  
  mean = summary_stats$mean_improvement,  
  lower = lower,  
  upper = upper  
)  
  
ggplot(mae_by_fold, aes(x = factor(fold), y = improvement)) +  
  geom_point(size = 3, color = "#2C7BB6") +  
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +  
  
  # Mean point  
  geom_point(  
    data = summary_df,  
    aes(x = x, y = mean),  
    inherit.aes = FALSE,  
    size = 5,  
    color = "black"  
  ) +  
  
  # Confidence interval  
  geom_errorbar(  
    data = summary_df,  
    aes(x = x, ymin = lower, ymax = upper),  
    inherit.aes = FALSE,  
    width = 0.2,  
    color = "black",  
    size = 1.2  
  ) +  
  
  labs(  
    title = "Model Improvement with Confidence Interval",  
    subtitle = "Confidence interval crosses zero, indicating uncertainty in improvement",  
    x = "Fold",  
    y = "Improvement (MAE_B - MAE_A)"  
  ) +  
  theme_minimal()
```


Model Improvement with Confidence Interval

Confidence interval crosses zero, indicating uncertainty in improvement



This section below is focusing on are we using the right fraud threshold

Step 1: Loading the data and perform data inspection

```
library(tidyverse)
df <- read.csv('workers_comp_fraud.csv')

glimpse(df)
```

Rows: 2,000

Columns: 3

```
$ claim_id    <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, ...
$ true_label  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ...
$ pred_prob   <dbl> 0.0002, 0.0326, 0.0001, 0.0188, 0.0006, 0.0007, 0.0186, 0.0...
```

Note: Cost rules-> If investigate: Not Fraud-> cost = \$500, fraud->cost = \$0. If not investigate: Fraud->cost = \$4,500, not fraud-> cost = 0.

Step 2: Write cost function (for one threshold)

```
threshold <- 0.10

df_cost <- df %>%
  mutate(
    decision = ifelse(pred_prob > threshold, "investigate", "no_investigate"),

    cost = case_when(
      decision == "investigate" & true_label == 0 ~ 500,
      decision == "investigate" & true_label == 1 ~ 0,
      decision == "no_investigate" & true_label == 1 ~ 4500,
      decision == "no_investigate" & true_label == 0 ~ 0
    )
  )

df_cost
```

	claim_id	true_label	pred_prob	decision	cost
1	1	0	0.0002	no_investigate	0
2	2	0	0.0326	no_investigate	0
3	3	0	0.0001	no_investigate	0
4	4	0	0.0188	no_investigate	0
5	5	0	0.0006	no_investigate	0
6	6	0	0.0007	no_investigate	0
7	7	0	0.0186	no_investigate	0
8	8	0	0.0025	no_investigate	0
9	9	0	0.0011	no_investigate	0
10	10	0	0.0084	no_investigate	0
11	11	0	0.0001	no_investigate	0
12	12	0	0.0226	no_investigate	0
13	13	0	0.0407	no_investigate	0
14	14	0	0.2307	investigate	500
15	15	0	0.0063	no_investigate	0
16	16	0	0.0128	no_investigate	0
17	17	0	0.0004	no_investigate	0
18	18	0	0.0002	no_investigate	0
19	19	0	0.0015	no_investigate	0
20	20	0	0.0007	no_investigate	0
21	21	0	0.0189	no_investigate	0
22	22	0	0.0013	no_investigate	0
23	23	0	0.0053	no_investigate	0
24	24	0	0.0348	no_investigate	0
25	25	0	0.0103	no_investigate	0
26	26	0	0.0003	no_investigate	0
27	27	0	0.0013	no_investigate	0
28	28	0	0.0016	no_investigate	0
29	29	0	0.0578	no_investigate	0
30	30	1	0.0730	no_investigate	4500

31	31	0	0.0001 no_investigate	0
32	32	0	0.0026 no_investigate	0
33	33	0	0.0067 no_investigate	0
34	34	0	0.0254 no_investigate	0
35	35	0	0.0008 no_investigate	0
36	36	0	0.0009 no_investigate	0
37	37	0	0.0003 no_investigate	0
38	38	0	0.0008 no_investigate	0
39	39	0	0.0008 no_investigate	0
40	40	0	0.0016 no_investigate	0
41	41	0	0.0118 no_investigate	0
42	42	0	0.0179 no_investigate	0
43	43	0	0.1018 investigate	500
44	44	0	0.0173 no_investigate	0
45	45	1	0.0654 no_investigate	4500
46	46	1	0.0635 no_investigate	4500
47	47	0	0.1124 investigate	500
48	48	0	0.0002 no_investigate	0
49	49	0	0.0030 no_investigate	0
50	50	0	0.0000 no_investigate	0
51	51	0	0.0943 no_investigate	0
52	52	0	0.0095 no_investigate	0
53	53	0	0.4065 investigate	500
54	54	0	0.0054 no_investigate	0
55	55	0	0.0168 no_investigate	0
56	56	0	0.0000 no_investigate	0
57	57	0	0.0021 no_investigate	0
58	58	0	0.0008 no_investigate	0
59	59	0	0.0006 no_investigate	0
60	60	1	0.0157 no_investigate	4500
61	61	0	0.0023 no_investigate	0
62	62	0	0.0021 no_investigate	0
63	63	0	0.0099 no_investigate	0
64	64	0	0.0001 no_investigate	0
65	65	1	0.2822 investigate	0
66	66	0	0.0012 no_investigate	0
67	67	0	0.2158 investigate	500
68	68	0	0.0056 no_investigate	0
69	69	0	0.0001 no_investigate	0
70	70	0	0.0076 no_investigate	0
71	71	0	0.0154 no_investigate	0
72	72	0	0.0013 no_investigate	0
73	73	0	0.0337 no_investigate	0
74	74	0	0.0005 no_investigate	0
75	75	0	0.0494 no_investigate	0
76	76	0	0.0033 no_investigate	0
77	77	1	0.0384 no_investigate	4500
78	78	0	0.0004 no_investigate	0
79	79	0	0.1891 investigate	500
80	80	0	0.0002 no_investigate	0
81	81	1	0.0088 no_investigate	4500

82	82	0	0.0036	no_investigate	0
83	83	0	0.0027	no_investigate	0
84	84	0	0.0547	no_investigate	0
85	85	0	0.0331	no_investigate	0
86	86	0	0.0442	no_investigate	0
87	87	0	0.0017	no_investigate	0
88	88	0	0.0377	no_investigate	0
89	89	0	0.0049	no_investigate	0
90	90	0	0.3948	investigate	500
91	91	0	0.0003	no_investigate	0
92	92	0	0.0039	no_investigate	0
93	93	0	0.0023	no_investigate	0
94	94	0	0.0004	no_investigate	0
95	95	0	0.0131	no_investigate	0
96	96	0	0.5964	investigate	500
97	97	0	0.0011	no_investigate	0
98	98	0	0.0003	no_investigate	0
99	99	0	0.0287	no_investigate	0
100	100	0	0.0384	no_investigate	0
101	101	0	0.0021	no_investigate	0
102	102	0	0.0001	no_investigate	0
103	103	0	0.0089	no_investigate	0
104	104	0	0.0091	no_investigate	0
105	105	0	0.0066	no_investigate	0
106	106	0	0.0094	no_investigate	0
107	107	0	0.0207	no_investigate	0
108	108	0	0.0082	no_investigate	0
109	109	0	0.0113	no_investigate	0
110	110	0	0.0054	no_investigate	0
111	111	0	0.0020	no_investigate	0
112	112	0	0.0001	no_investigate	0
113	113	0	0.0002	no_investigate	0
114	114	0	0.0012	no_investigate	0
115	115	0	0.0076	no_investigate	0
116	116	0	0.0132	no_investigate	0
117	117	0	0.2020	investigate	500
118	118	1	0.2332	investigate	0
119	119	0	0.0002	no_investigate	0
120	120	0	0.3023	investigate	500
121	121	0	0.0039	no_investigate	0
122	122	0	0.0001	no_investigate	0
123	123	0	0.0000	no_investigate	0
124	124	1	0.0252	no_investigate	4500
125	125	0	0.3573	investigate	500
126	126	0	0.1679	investigate	500
127	127	0	0.0005	no_investigate	0
128	128	0	0.1026	investigate	500
129	129	0	0.0000	no_investigate	0
130	130	0	0.0019	no_investigate	0
131	131	0	0.0023	no_investigate	0
132	132	0	0.0027	no_investigate	0

133	133	0	0.0036	no_investigate	0
134	134	0	0.0139	no_investigate	0
135	135	0	0.0001	no_investigate	0
136	136	0	0.0009	no_investigate	0
137	137	0	0.0020	no_investigate	0
138	138	0	0.0723	no_investigate	0
139	139	0	0.0016	no_investigate	0
140	140	0	0.1996	investigate	500
141	141	0	0.3760	investigate	500
142	142	0	0.2773	investigate	500
143	143	0	0.0013	no_investigate	0
144	144	0	0.0026	no_investigate	0
145	145	0	0.0001	no_investigate	0
146	146	0	0.0001	no_investigate	0
147	147	0	0.0020	no_investigate	0
148	148	0	0.0525	no_investigate	0
149	149	0	0.0090	no_investigate	0
150	150	0	0.0007	no_investigate	0
151	151	0	0.0001	no_investigate	0
152	152	0	0.0005	no_investigate	0
153	153	0	0.0031	no_investigate	0
154	154	0	0.0011	no_investigate	0
155	155	1	0.3088	investigate	0
156	156	0	0.0159	no_investigate	0
157	157	0	0.1448	investigate	500
158	158	0	0.0496	no_investigate	0
159	159	0	0.0010	no_investigate	0
160	160	0	0.0027	no_investigate	0
161	161	0	0.0035	no_investigate	0
162	162	0	0.0482	no_investigate	0
163	163	0	0.1803	investigate	500
164	164	1	0.2490	investigate	0
165	165	0	0.0206	no_investigate	0
166	166	0	0.0048	no_investigate	0
167	167	0	0.0000	no_investigate	0
168	168	0	0.0042	no_investigate	0
169	169	0	0.0027	no_investigate	0
170	170	0	0.0029	no_investigate	0
171	171	0	0.0004	no_investigate	0
172	172	0	0.0012	no_investigate	0
173	173	0	0.0462	no_investigate	0
174	174	0	0.0063	no_investigate	0
175	175	0	0.0005	no_investigate	0
176	176	0	0.0606	no_investigate	0
177	177	0	0.0757	no_investigate	0
178	178	0	0.0030	no_investigate	0
179	179	0	0.0522	no_investigate	0
180	180	0	0.0000	no_investigate	0
181	181	0	0.1131	investigate	500
182	182	1	0.4835	investigate	0
183	183	0	0.0688	no_investigate	0

184	184	0	0.0010	no_investigate	0
185	185	0	0.0015	no_investigate	0
186	186	0	0.0175	no_investigate	0
187	187	0	0.0522	no_investigate	0
188	188	0	0.0007	no_investigate	0
189	189	0	0.0568	no_investigate	0
190	190	0	0.0011	no_investigate	0
191	191	0	0.0140	no_investigate	0
192	192	0	0.0001	no_investigate	0
193	193	0	0.0006	no_investigate	0
194	194	0	0.0001	no_investigate	0
195	195	0	0.2435	investigate	500
196	196	0	0.1209	investigate	500
197	197	0	0.0728	no_investigate	0
198	198	0	0.0034	no_investigate	0
199	199	0	0.0011	no_investigate	0
200	200	1	0.1715	investigate	0
201	201	0	0.0036	no_investigate	0
202	202	0	0.0092	no_investigate	0
203	203	0	0.1213	investigate	500
204	204	0	0.0199	no_investigate	0
205	205	0	0.0001	no_investigate	0
206	206	0	0.0256	no_investigate	0
207	207	0	0.0014	no_investigate	0
208	208	0	0.0085	no_investigate	0
209	209	1	0.0149	no_investigate	4500
210	210	0	0.0046	no_investigate	0
211	211	0	0.0337	no_investigate	0
212	212	0	0.0017	no_investigate	0
213	213	0	0.1017	investigate	500
214	214	0	0.0490	no_investigate	0
215	215	0	0.0081	no_investigate	0
216	216	0	0.0490	no_investigate	0
217	217	0	0.0632	no_investigate	0
218	218	0	0.3100	investigate	500
219	219	0	0.0029	no_investigate	0
220	220	0	0.0079	no_investigate	0
221	221	0	0.0034	no_investigate	0
222	222	0	0.0356	no_investigate	0
223	223	0	0.0821	no_investigate	0
224	224	0	0.0003	no_investigate	0
225	225	0	0.0020	no_investigate	0
226	226	0	0.0001	no_investigate	0
227	227	0	0.0941	no_investigate	0
228	228	0	0.0038	no_investigate	0
229	229	0	0.0015	no_investigate	0
230	230	0	0.0007	no_investigate	0
231	231	0	0.0264	no_investigate	0
232	232	0	0.0077	no_investigate	0
233	233	0	0.0211	no_investigate	0
234	234	0	0.0017	no_investigate	0

235	235	0	0.0002	no_investigate	0
236	236	0	0.0137	no_investigate	0
237	237	1	0.6514	investigate	0
238	238	0	0.0035	no_investigate	0
239	239	0	0.0032	no_investigate	0
240	240	0	0.0011	no_investigate	0
241	241	0	0.0385	no_investigate	0
242	242	1	0.3759	investigate	0
243	243	0	0.0026	no_investigate	0
244	244	0	0.0005	no_investigate	0
245	245	0	0.0049	no_investigate	0
246	246	0	0.0083	no_investigate	0
247	247	0	0.0296	no_investigate	0
248	248	0	0.0198	no_investigate	0
249	249	0	0.0025	no_investigate	0
250	250	0	0.0124	no_investigate	0
251	251	0	0.0027	no_investigate	0
252	252	0	0.0000	no_investigate	0
253	253	0	0.0007	no_investigate	0
254	254	0	0.0071	no_investigate	0
255	255	0	0.0003	no_investigate	0
256	256	0	0.0266	no_investigate	0
257	257	0	0.0007	no_investigate	0
258	258	0	0.0003	no_investigate	0
259	259	0	0.0001	no_investigate	0
260	260	0	0.0004	no_investigate	0
261	261	0	0.0134	no_investigate	0
262	262	0	0.0012	no_investigate	0
263	263	0	0.0047	no_investigate	0
264	264	0	0.0220	no_investigate	0
265	265	0	0.0088	no_investigate	0
266	266	0	0.0003	no_investigate	0
267	267	0	0.5496	investigate	500
268	268	0	0.0009	no_investigate	0
269	269	1	0.1227	investigate	0
270	270	0	0.0022	no_investigate	0
271	271	0	0.0001	no_investigate	0
272	272	0	0.0218	no_investigate	0
273	273	0	0.0673	no_investigate	0
274	274	0	0.0178	no_investigate	0
275	275	1	0.2059	investigate	0
276	276	0	0.2581	investigate	500
277	277	0	0.0161	no_investigate	0
278	278	0	0.1575	investigate	500
279	279	0	0.0753	no_investigate	0
280	280	0	0.0005	no_investigate	0
281	281	0	0.0106	no_investigate	0
282	282	0	0.0003	no_investigate	0
283	283	0	0.0223	no_investigate	0
284	284	0	0.0011	no_investigate	0
285	285	0	0.0181	no_investigate	0

286	286	0	0.0471	no_investigate	0
287	287	0	0.0044	no_investigate	0
288	288	0	0.0036	no_investigate	0
289	289	0	0.0083	no_investigate	0
290	290	0	0.0425	no_investigate	0
291	291	0	0.0008	no_investigate	0
292	292	0	0.0010	no_investigate	0
293	293	0	0.0005	no_investigate	0
294	294	0	0.0005	no_investigate	0
295	295	0	0.0027	no_investigate	0
296	296	0	0.0630	no_investigate	0
297	297	0	0.0001	no_investigate	0
298	298	0	0.0004	no_investigate	0
299	299	0	0.0686	no_investigate	0
300	300	0	0.0527	no_investigate	0
301	301	1	0.0624	no_investigate	4500
302	302	0	0.0033	no_investigate	0
303	303	0	0.0096	no_investigate	0
304	304	0	0.0002	no_investigate	0
305	305	0	0.0035	no_investigate	0
306	306	0	0.0085	no_investigate	0
307	307	0	0.0145	no_investigate	0
308	308	0	0.0107	no_investigate	0
309	309	0	0.0193	no_investigate	0
310	310	0	0.0028	no_investigate	0
311	311	0	0.0092	no_investigate	0
312	312	0	0.0064	no_investigate	0
313	313	0	0.0005	no_investigate	0
314	314	0	0.0401	no_investigate	0
315	315	0	0.0148	no_investigate	0
316	316	0	0.0030	no_investigate	0
317	317	0	0.0114	no_investigate	0
318	318	0	0.0361	no_investigate	0
319	319	1	0.0128	no_investigate	4500
320	320	0	0.0013	no_investigate	0
321	321	0	0.0002	no_investigate	0
322	322	0	0.0012	no_investigate	0
323	323	1	0.7607	investigate	0
324	324	0	0.0267	no_investigate	0
325	325	0	0.0000	no_investigate	0
326	326	0	0.0005	no_investigate	0
327	327	0	0.0002	no_investigate	0
328	328	0	0.0106	no_investigate	0
329	329	0	0.0008	no_investigate	0
330	330	0	0.0546	no_investigate	0
331	331	0	0.0008	no_investigate	0
332	332	0	0.0042	no_investigate	0
333	333	0	0.2794	investigate	500
334	334	0	0.0008	no_investigate	0
335	335	0	0.0065	no_investigate	0
336	336	0	0.0001	no_investigate	0

337	337	0	0.0110	no_investigate	0
338	338	0	0.0105	no_investigate	0
339	339	0	0.0015	no_investigate	0
340	340	0	0.0320	no_investigate	0
341	341	0	0.0130	no_investigate	0
342	342	0	0.0045	no_investigate	0
343	343	0	0.0090	no_investigate	0
344	344	0	0.0909	no_investigate	0
345	345	0	0.1818	investigate	500
346	346	0	0.1021	investigate	500
347	347	0	0.0093	no_investigate	0
348	348	1	0.2010	investigate	0
349	349	0	0.0031	no_investigate	0
350	350	0	0.6180	investigate	500
351	351	0	0.0620	no_investigate	0
352	352	0	0.0304	no_investigate	0
353	353	0	0.0419	no_investigate	0
354	354	0	0.0396	no_investigate	0
355	355	0	0.0045	no_investigate	0
356	356	0	0.0016	no_investigate	0
357	357	0	0.1236	investigate	500
358	358	0	0.0015	no_investigate	0
359	359	0	0.0445	no_investigate	0
360	360	0	0.0248	no_investigate	0
361	361	0	0.0014	no_investigate	0
362	362	0	0.0005	no_investigate	0
363	363	0	0.0117	no_investigate	0
364	364	0	0.0097	no_investigate	0
365	365	0	0.0038	no_investigate	0
366	366	0	0.0237	no_investigate	0
367	367	0	0.0005	no_investigate	0
368	368	0	0.0003	no_investigate	0
369	369	0	0.0031	no_investigate	0
370	370	0	0.0245	no_investigate	0
371	371	0	0.0821	no_investigate	0
372	372	1	0.1247	investigate	0
373	373	0	0.0068	no_investigate	0
374	374	0	0.0001	no_investigate	0
375	375	0	0.0453	no_investigate	0
376	376	0	0.6133	investigate	500
377	377	0	0.0287	no_investigate	0
378	378	0	0.1465	investigate	500
379	379	1	0.2032	investigate	0
380	380	0	0.0068	no_investigate	0
381	381	0	0.0558	no_investigate	0
382	382	0	0.0077	no_investigate	0
383	383	0	0.1008	investigate	500
384	384	0	0.0132	no_investigate	0
385	385	0	0.0001	no_investigate	0
386	386	0	0.0015	no_investigate	0
387	387	0	0.0006	no_investigate	0

388	388	1	0.2066	investigate	0
389	389	0	0.0023	no_investigate	0
390	390	0	0.0306	no_investigate	0
391	391	0	0.0002	no_investigate	0
392	392	0	0.0028	no_investigate	0
393	393	0	0.0047	no_investigate	0
394	394	0	0.0013	no_investigate	0
395	395	0	0.0002	no_investigate	0
396	396	0	0.0076	no_investigate	0
397	397	0	0.0011	no_investigate	0
398	398	0	0.0005	no_investigate	0
399	399	0	0.0018	no_investigate	0
400	400	0	0.0018	no_investigate	0
401	401	0	0.0224	no_investigate	0
402	402	0	0.0039	no_investigate	0
403	403	0	0.0067	no_investigate	0
404	404	0	0.0009	no_investigate	0
405	405	0	0.0168	no_investigate	0
406	406	0	0.0093	no_investigate	0
407	407	0	0.0027	no_investigate	0
408	408	0	0.1102	investigate	500
409	409	0	0.0008	no_investigate	0
410	410	0	0.0066	no_investigate	0
411	411	1	0.0014	no_investigate	4500
412	412	0	0.0003	no_investigate	0
413	413	1	0.7363	investigate	0
414	414	0	0.0009	no_investigate	0
415	415	0	0.0040	no_investigate	0
416	416	0	0.0005	no_investigate	0
417	417	0	0.0176	no_investigate	0
418	418	0	0.0012	no_investigate	0
419	419	0	0.0008	no_investigate	0
420	420	0	0.0006	no_investigate	0
421	421	0	0.0042	no_investigate	0
422	422	0	0.0379	no_investigate	0
423	423	0	0.0001	no_investigate	0
424	424	0	0.0008	no_investigate	0
425	425	0	0.0008	no_investigate	0
426	426	0	0.0011	no_investigate	0
427	427	1	0.0144	no_investigate	4500
428	428	0	0.0173	no_investigate	0
429	429	0	0.0003	no_investigate	0
430	430	0	0.2027	investigate	500
431	431	0	0.0011	no_investigate	0
432	432	0	0.0002	no_investigate	0
433	433	0	0.0048	no_investigate	0
434	434	0	0.0179	no_investigate	0
435	435	0	0.0089	no_investigate	0
436	436	0	0.0758	no_investigate	0
437	437	0	0.0256	no_investigate	0
438	438	0	0.0288	no_investigate	0

439	439	0	0.0200	no_investigate	0
440	440	0	0.0007	no_investigate	0
441	441	0	0.0070	no_investigate	0
442	442	1	0.0036	no_investigate	4500
443	443	0	0.0097	no_investigate	0
444	444	0	0.0000	no_investigate	0
445	445	0	0.0049	no_investigate	0
446	446	0	0.5994	investigate	500
447	447	0	0.0003	no_investigate	0
448	448	0	0.0005	no_investigate	0
449	449	0	0.0004	no_investigate	0
450	450	0	0.0023	no_investigate	0
451	451	0	0.0008	no_investigate	0
452	452	1	0.0265	no_investigate	4500
453	453	0	0.0764	no_investigate	0
454	454	0	0.0004	no_investigate	0
455	455	0	0.0131	no_investigate	0
456	456	0	0.0182	no_investigate	0
457	457	0	0.0000	no_investigate	0
458	458	0	0.0023	no_investigate	0
459	459	0	0.0004	no_investigate	0
460	460	0	0.0050	no_investigate	0
461	461	0	0.0006	no_investigate	0
462	462	0	0.0023	no_investigate	0
463	463	0	0.0057	no_investigate	0
464	464	0	0.0031	no_investigate	0
465	465	0	0.0001	no_investigate	0
466	466	0	0.0045	no_investigate	0
467	467	0	0.0038	no_investigate	0
468	468	0	0.0247	no_investigate	0
469	469	0	0.0371	no_investigate	0
470	470	0	0.0103	no_investigate	0
471	471	0	0.0000	no_investigate	0
472	472	0	0.0001	no_investigate	0
473	473	0	0.0094	no_investigate	0
474	474	0	0.0121	no_investigate	0
475	475	0	0.0123	no_investigate	0
476	476	0	0.0139	no_investigate	0
477	477	0	0.0161	no_investigate	0
478	478	0	0.0005	no_investigate	0
479	479	1	0.9375	investigate	0
480	480	0	0.0020	no_investigate	0
481	481	0	0.0073	no_investigate	0
482	482	0	0.0001	no_investigate	0
483	483	0	0.0195	no_investigate	0
484	484	0	0.0089	no_investigate	0
485	485	0	0.0006	no_investigate	0
486	486	0	0.0881	no_investigate	0
487	487	0	0.0026	no_investigate	0
488	488	0	0.0007	no_investigate	0
489	489	0	0.0028	no_investigate	0

490	490	0	0.0002	no_investigate	0
491	491	1	0.4445	investigate	0
492	492	0	0.0678	no_investigate	0
493	493	0	0.0023	no_investigate	0
494	494	0	0.0134	no_investigate	0
495	495	0	0.0001	no_investigate	0
496	496	0	0.0328	no_investigate	0
497	497	1	0.2289	investigate	0
498	498	0	0.0015	no_investigate	0
499	499	0	0.0374	no_investigate	0
500	500	0	0.0026	no_investigate	0
501	501	0	0.0013	no_investigate	0
502	502	0	0.0016	no_investigate	0
503	503	0	0.0135	no_investigate	0
504	504	0	0.0252	no_investigate	0
505	505	0	0.0057	no_investigate	0
506	506	0	0.0040	no_investigate	0
507	507	0	0.0090	no_investigate	0
508	508	0	0.0158	no_investigate	0
509	509	0	0.0007	no_investigate	0
510	510	0	0.0002	no_investigate	0
511	511	0	0.0255	no_investigate	0
512	512	0	0.0001	no_investigate	0
513	513	0	0.0009	no_investigate	0
514	514	0	0.0038	no_investigate	0
515	515	0	0.2386	investigate	500
516	516	0	0.0005	no_investigate	0
517	517	0	0.0014	no_investigate	0
518	518	0	0.0011	no_investigate	0
519	519	0	0.0031	no_investigate	0
520	520	0	0.0073	no_investigate	0
521	521	0	0.0134	no_investigate	0
522	522	0	0.0127	no_investigate	0
523	523	0	0.0120	no_investigate	0
524	524	0	0.0092	no_investigate	0
525	525	0	0.0165	no_investigate	0
526	526	1	0.0101	no_investigate	4500
527	527	0	0.0006	no_investigate	0
528	528	0	0.0003	no_investigate	0
529	529	0	0.0658	no_investigate	0
530	530	0	0.0005	no_investigate	0
531	531	0	0.0147	no_investigate	0
532	532	0	0.0002	no_investigate	0
533	533	0	0.0008	no_investigate	0
534	534	0	0.0007	no_investigate	0
535	535	0	0.1356	investigate	500
536	536	0	0.0012	no_investigate	0
537	537	0	0.0066	no_investigate	0
538	538	0	0.1208	investigate	500
539	539	0	0.0025	no_investigate	0
540	540	1	0.0190	no_investigate	4500

541	541	0	0.0285	no_investigate	0
542	542	0	0.0005	no_investigate	0
543	543	1	0.5978	investigate	0
544	544	0	0.0244	no_investigate	0
545	545	0	0.0288	no_investigate	0
546	546	0	0.0031	no_investigate	0
547	547	0	0.0040	no_investigate	0
548	548	0	0.0553	no_investigate	0
549	549	0	0.0034	no_investigate	0
550	550	0	0.0006	no_investigate	0
551	551	1	0.0050	no_investigate	4500
552	552	0	0.0083	no_investigate	0
553	553	0	0.0031	no_investigate	0
554	554	0	0.0048	no_investigate	0
555	555	0	0.0200	no_investigate	0
556	556	0	0.0024	no_investigate	0
557	557	0	0.0037	no_investigate	0
558	558	0	0.0041	no_investigate	0
559	559	1	0.1204	investigate	0
560	560	0	0.1331	investigate	500
561	561	0	0.0000	no_investigate	0
562	562	0	0.0034	no_investigate	0
563	563	0	0.0275	no_investigate	0
564	564	0	0.0001	no_investigate	0
565	565	1	0.5517	investigate	0
566	566	0	0.1377	investigate	500
567	567	0	0.0001	no_investigate	0
568	568	0	0.0152	no_investigate	0
569	569	0	0.0007	no_investigate	0
570	570	0	0.0366	no_investigate	0
571	571	0	0.0224	no_investigate	0
572	572	0	0.0312	no_investigate	0
573	573	0	0.2230	investigate	500
574	574	0	0.0002	no_investigate	0
575	575	0	0.0205	no_investigate	0
576	576	0	0.0273	no_investigate	0
577	577	0	0.0066	no_investigate	0
578	578	0	0.0005	no_investigate	0
579	579	0	0.0015	no_investigate	0
580	580	0	0.0060	no_investigate	0
581	581	0	0.0021	no_investigate	0
582	582	1	0.0506	no_investigate	4500
583	583	1	0.0718	no_investigate	4500
584	584	0	0.1001	investigate	500
585	585	0	0.0010	no_investigate	0
586	586	0	0.0026	no_investigate	0
587	587	0	0.0063	no_investigate	0
588	588	0	0.0885	no_investigate	0
589	589	1	0.0006	no_investigate	4500
590	590	0	0.0007	no_investigate	0
591	591	0	0.0066	no_investigate	0

592	592	0	0.0027	no_investigate	0
593	593	0	0.0134	no_investigate	0
594	594	0	0.0039	no_investigate	0
595	595	0	0.1293	investigate	500
596	596	0	0.0011	no_investigate	0
597	597	0	0.0221	no_investigate	0
598	598	1	0.0389	no_investigate	4500
599	599	0	0.0194	no_investigate	0
600	600	0	0.0005	no_investigate	0
601	601	0	0.0365	no_investigate	0
602	602	0	0.0327	no_investigate	0
603	603	0	0.0002	no_investigate	0
604	604	0	0.0003	no_investigate	0
605	605	0	0.0001	no_investigate	0
606	606	0	0.0081	no_investigate	0
607	607	0	0.0005	no_investigate	0
608	608	0	0.0402	no_investigate	0
609	609	1	0.4847	investigate	0
610	610	0	0.0223	no_investigate	0
611	611	0	0.1487	investigate	500
612	612	0	0.0080	no_investigate	0
613	613	0	0.0006	no_investigate	0
614	614	0	0.0012	no_investigate	0
615	615	0	0.0028	no_investigate	0
616	616	0	0.0007	no_investigate	0
617	617	0	0.0003	no_investigate	0
618	618	0	0.0003	no_investigate	0
619	619	0	0.1585	investigate	500
620	620	0	0.0312	no_investigate	0
621	621	0	0.0001	no_investigate	0
622	622	0	0.0005	no_investigate	0
623	623	0	0.0065	no_investigate	0
624	624	0	0.0013	no_investigate	0
625	625	0	0.0267	no_investigate	0
626	626	0	0.0082	no_investigate	0
627	627	0	0.0006	no_investigate	0
628	628	0	0.0005	no_investigate	0
629	629	0	0.0002	no_investigate	0
630	630	0	0.0046	no_investigate	0
631	631	0	0.0050	no_investigate	0
632	632	0	0.0182	no_investigate	0
633	633	0	0.0112	no_investigate	0
634	634	0	0.0069	no_investigate	0
635	635	0	0.0042	no_investigate	0
636	636	0	0.0018	no_investigate	0
637	637	0	0.0003	no_investigate	0
638	638	0	0.0469	no_investigate	0
639	639	0	0.0052	no_investigate	0
640	640	0	0.0001	no_investigate	0
641	641	0	0.0022	no_investigate	0
642	642	0	0.0003	no_investigate	0

643	643	0	0.0003	no_investigate	0
644	644	0	0.0049	no_investigate	0
645	645	0	0.0255	no_investigate	0
646	646	0	0.0166	no_investigate	0
647	647	1	0.8638	investigate	0
648	648	0	0.0728	no_investigate	0
649	649	0	0.0047	no_investigate	0
650	650	0	0.0033	no_investigate	0
651	651	1	0.0744	no_investigate	4500
652	652	0	0.0105	no_investigate	0
653	653	0	0.0141	no_investigate	0
654	654	0	0.0114	no_investigate	0
655	655	0	0.0001	no_investigate	0
656	656	0	0.0303	no_investigate	0
657	657	0	0.0828	no_investigate	0
658	658	0	0.0038	no_investigate	0
659	659	0	0.0084	no_investigate	0
660	660	1	0.0031	no_investigate	4500
661	661	0	0.0378	no_investigate	0
662	662	0	0.0013	no_investigate	0
663	663	0	0.0523	no_investigate	0
664	664	0	0.0001	no_investigate	0
665	665	0	0.0046	no_investigate	0
666	666	0	0.0779	no_investigate	0
667	667	0	0.0696	no_investigate	0
668	668	0	0.0100	no_investigate	0
669	669	0	0.0003	no_investigate	0
670	670	0	0.0010	no_investigate	0
671	671	0	0.0014	no_investigate	0
672	672	0	0.0047	no_investigate	0
673	673	0	0.0044	no_investigate	0
674	674	0	0.0002	no_investigate	0
675	675	0	0.0007	no_investigate	0
676	676	0	0.0725	no_investigate	0
677	677	0	0.0000	no_investigate	0
678	678	0	0.0102	no_investigate	0
679	679	0	0.0209	no_investigate	0
680	680	0	0.0111	no_investigate	0
681	681	0	0.0088	no_investigate	0
682	682	1	0.0187	no_investigate	4500
683	683	0	0.0135	no_investigate	0
684	684	0	0.0005	no_investigate	0
685	685	0	0.0000	no_investigate	0
686	686	0	0.0049	no_investigate	0
687	687	0	0.0001	no_investigate	0
688	688	0	0.6091	investigate	500
689	689	0	0.0049	no_investigate	0
690	690	0	0.0013	no_investigate	0
691	691	0	0.0029	no_investigate	0
692	692	0	0.0072	no_investigate	0
693	693	0	0.0026	no_investigate	0

694	694	1	0.0013	no_investigate	4500
695	695	0	0.0089	no_investigate	0
696	696	0	0.0001	no_investigate	0
697	697	0	0.0001	no_investigate	0
698	698	0	0.0030	no_investigate	0
699	699	0	0.0564	no_investigate	0
700	700	0	0.0123	no_investigate	0
701	701	0	0.0007	no_investigate	0
702	702	0	0.0002	no_investigate	0
703	703	0	0.0002	no_investigate	0
704	704	0	0.0003	no_investigate	0
705	705	0	0.0000	no_investigate	0
706	706	0	0.0162	no_investigate	0
707	707	0	0.0145	no_investigate	0
708	708	0	0.0023	no_investigate	0
709	709	0	0.0015	no_investigate	0
710	710	0	0.1706	investigate	500
711	711	0	0.0004	no_investigate	0
712	712	0	0.0474	no_investigate	0
713	713	0	0.0089	no_investigate	0
714	714	0	0.0000	no_investigate	0
715	715	0	0.0099	no_investigate	0
716	716	0	0.0781	no_investigate	0
717	717	0	0.0875	no_investigate	0
718	718	0	0.0016	no_investigate	0
719	719	0	0.0016	no_investigate	0
720	720	0	0.0001	no_investigate	0
721	721	0	0.1290	investigate	500
722	722	1	0.7849	investigate	0
723	723	0	0.0031	no_investigate	0
724	724	0	0.0001	no_investigate	0
725	725	0	0.0252	no_investigate	0
726	726	0	0.0035	no_investigate	0
727	727	0	0.0030	no_investigate	0
728	728	0	0.5497	investigate	500
729	729	1	0.1049	investigate	0
730	730	0	0.0011	no_investigate	0
731	731	0	0.1401	investigate	500
732	732	0	0.5355	investigate	500
733	733	0	0.0127	no_investigate	0
734	734	0	0.0068	no_investigate	0
735	735	0	0.0115	no_investigate	0
736	736	0	0.0044	no_investigate	0
737	737	0	0.0001	no_investigate	0
738	738	0	0.0023	no_investigate	0
739	739	0	0.0959	no_investigate	0
740	740	0	0.0468	no_investigate	0
741	741	0	0.0002	no_investigate	0
742	742	0	0.0009	no_investigate	0
743	743	0	0.0022	no_investigate	0
744	744	0	0.0000	no_investigate	0

745	745	0	0.0143	no_investigate	0
746	746	0	0.0008	no_investigate	0
747	747	0	0.0080	no_investigate	0
748	748	1	0.8270	investigate	0
749	749	0	0.0434	no_investigate	0
750	750	0	0.0056	no_investigate	0
751	751	0	0.0001	no_investigate	0
752	752	0	0.0027	no_investigate	0
753	753	0	0.0120	no_investigate	0
754	754	0	0.0029	no_investigate	0
755	755	0	0.1079	investigate	500
756	756	0	0.0059	no_investigate	0
757	757	1	0.1082	investigate	0
758	758	0	0.0262	no_investigate	0
759	759	0	0.0037	no_investigate	0
760	760	0	0.0016	no_investigate	0
761	761	0	0.0031	no_investigate	0
762	762	0	0.0005	no_investigate	0
763	763	0	0.0182	no_investigate	0
764	764	0	0.0818	no_investigate	0
765	765	0	0.0202	no_investigate	0
766	766	0	0.0012	no_investigate	0
767	767	1	0.1047	investigate	0
768	768	1	0.2201	investigate	0
769	769	0	0.0876	no_investigate	0
770	770	0	0.3409	investigate	500
771	771	0	0.0012	no_investigate	0
772	772	0	0.0441	no_investigate	0
773	773	0	0.0000	no_investigate	0
774	774	0	0.0003	no_investigate	0
775	775	0	0.0147	no_investigate	0
776	776	0	0.0083	no_investigate	0
777	777	0	0.0088	no_investigate	0
778	778	0	0.0444	no_investigate	0
779	779	0	0.0221	no_investigate	0
780	780	0	0.0211	no_investigate	0
781	781	0	0.0035	no_investigate	0
782	782	0	0.0027	no_investigate	0
783	783	1	0.0245	no_investigate	4500
784	784	0	0.0047	no_investigate	0
785	785	0	0.0020	no_investigate	0
786	786	0	0.4175	investigate	500
787	787	0	0.0312	no_investigate	0
788	788	0	0.0022	no_investigate	0
789	789	1	0.0061	no_investigate	4500
790	790	0	0.0054	no_investigate	0
791	791	0	0.0028	no_investigate	0
792	792	0	0.0022	no_investigate	0
793	793	1	0.0074	no_investigate	4500
794	794	0	0.2234	investigate	500
795	795	0	0.0004	no_investigate	0

796	796	0	0.0017 no_investigate	0
797	797	0	0.0099 no_investigate	0
798	798	0	0.0058 no_investigate	0
799	799	1	0.0038 no_investigate	4500
800	800	0	0.0703 no_investigate	0
801	801	0	0.0007 no_investigate	0
802	802	0	0.0001 no_investigate	0
803	803	0	0.0057 no_investigate	0
804	804	0	0.0021 no_investigate	0
805	805	0	0.0131 no_investigate	0
806	806	0	0.0007 no_investigate	0
807	807	0	0.0148 no_investigate	0
808	808	0	0.1478 investigate	500
809	809	0	0.0001 no_investigate	0
810	810	0	0.0294 no_investigate	0
811	811	0	0.0034 no_investigate	0
812	812	0	0.0012 no_investigate	0
813	813	0	0.0088 no_investigate	0
814	814	0	0.0015 no_investigate	0
815	815	0	0.0014 no_investigate	0
816	816	0	0.0001 no_investigate	0
817	817	1	0.4758 investigate	0
818	818	0	0.0006 no_investigate	0
819	819	1	0.4786 investigate	0
820	820	0	0.0092 no_investigate	0
821	821	0	0.0019 no_investigate	0
822	822	0	0.0001 no_investigate	0
823	823	0	0.0461 no_investigate	0
824	824	0	0.0096 no_investigate	0
825	825	0	0.2593 investigate	500
826	826	0	0.0025 no_investigate	0
827	827	0	0.0054 no_investigate	0
828	828	0	0.0008 no_investigate	0
829	829	0	0.0283 no_investigate	0
830	830	0	0.0022 no_investigate	0
831	831	0	0.0100 no_investigate	0
832	832	0	0.0072 no_investigate	0
833	833	0	0.0003 no_investigate	0
834	834	0	0.0204 no_investigate	0
835	835	0	0.0009 no_investigate	0
836	836	1	0.0072 no_investigate	4500
837	837	0	0.0062 no_investigate	0
838	838	0	0.0009 no_investigate	0
839	839	0	0.0043 no_investigate	0
840	840	1	0.0295 no_investigate	4500
841	841	0	0.0133 no_investigate	0
842	842	0	0.0023 no_investigate	0
843	843	0	0.0047 no_investigate	0
844	844	0	0.0016 no_investigate	0
845	845	0	0.4302 investigate	500
846	846	0	0.0207 no_investigate	0

847	847	0	0.0001 no_investigate	0
848	848	0	0.0093 no_investigate	0
849	849	0	0.0000 no_investigate	0
850	850	0	0.0028 no_investigate	0
851	851	0	0.0052 no_investigate	0
852	852	0	0.0072 no_investigate	0
853	853	0	0.0030 no_investigate	0
854	854	1	0.0463 no_investigate	4500
855	855	0	0.0014 no_investigate	0
856	856	1	0.0142 no_investigate	4500
857	857	0	0.0070 no_investigate	0
858	858	0	0.0038 no_investigate	0
859	859	0	0.0647 no_investigate	0
860	860	0	0.0292 no_investigate	0
861	861	0	0.1081 investigate	500
862	862	0	0.0094 no_investigate	0
863	863	0	0.0279 no_investigate	0
864	864	0	0.0008 no_investigate	0
865	865	0	0.0018 no_investigate	0
866	866	0	0.0007 no_investigate	0
867	867	0	0.0050 no_investigate	0
868	868	0	0.1332 investigate	500
869	869	0	0.0022 no_investigate	0
870	870	0	0.0020 no_investigate	0
871	871	0	0.0086 no_investigate	0
872	872	0	0.0010 no_investigate	0
873	873	0	0.0004 no_investigate	0
874	874	1	0.0501 no_investigate	4500
875	875	0	0.0138 no_investigate	0
876	876	1	0.1675 investigate	0
877	877	0	0.0026 no_investigate	0
878	878	0	0.0229 no_investigate	0
879	879	0	0.0608 no_investigate	0
880	880	0	0.0027 no_investigate	0
881	881	0	0.0741 no_investigate	0
882	882	0	0.0024 no_investigate	0
883	883	0	0.0000 no_investigate	0
884	884	0	0.0009 no_investigate	0
885	885	0	0.0005 no_investigate	0
886	886	0	0.4072 investigate	500
887	887	0	0.0074 no_investigate	0
888	888	0	0.0197 no_investigate	0
889	889	0	0.1682 investigate	500
890	890	0	0.0014 no_investigate	0
891	891	1	0.0037 no_investigate	4500
892	892	0	0.0437 no_investigate	0
893	893	0	0.0000 no_investigate	0
894	894	0	0.0005 no_investigate	0
895	895	0	0.0130 no_investigate	0
896	896	0	0.0018 no_investigate	0
897	897	1	0.8991 investigate	0

898	898	0	0.0048	no_investigate	0
899	899	0	0.0000	no_investigate	0
900	900	1	0.0065	no_investigate	4500
901	901	0	0.0058	no_investigate	0
902	902	0	0.0641	no_investigate	0
903	903	0	0.0019	no_investigate	0
904	904	0	0.0149	no_investigate	0
905	905	0	0.0088	no_investigate	0
906	906	0	0.0706	no_investigate	0
907	907	0	0.0005	no_investigate	0
908	908	0	0.0051	no_investigate	0
909	909	0	0.0000	no_investigate	0
910	910	0	0.0029	no_investigate	0
911	911	0	0.0004	no_investigate	0
912	912	0	0.0004	no_investigate	0
913	913	0	0.0001	no_investigate	0
914	914	0	0.0032	no_investigate	0
915	915	0	0.0761	no_investigate	0
916	916	0	0.0014	no_investigate	0
917	917	0	0.0046	no_investigate	0
918	918	0	0.0014	no_investigate	0
919	919	0	0.0066	no_investigate	0
920	920	0	0.0029	no_investigate	0
921	921	0	0.0013	no_investigate	0
922	922	0	0.0022	no_investigate	0
923	923	0	0.0263	no_investigate	0
924	924	1	0.0001	no_investigate	4500
925	925	0	0.0358	no_investigate	0
926	926	0	0.0004	no_investigate	0
927	927	0	0.0012	no_investigate	0
928	928	0	0.5413	investigate	500
929	929	0	0.0002	no_investigate	0
930	930	0	0.0079	no_investigate	0
931	931	0	0.0147	no_investigate	0
932	932	0	0.5055	investigate	500
933	933	0	0.1533	investigate	500
934	934	0	0.0307	no_investigate	0
935	935	0	0.1521	investigate	500
936	936	0	0.0008	no_investigate	0
937	937	1	0.0055	no_investigate	4500
938	938	0	0.0006	no_investigate	0
939	939	0	0.0026	no_investigate	0
940	940	0	0.0000	no_investigate	0
941	941	0	0.0007	no_investigate	0
942	942	0	0.0378	no_investigate	0
943	943	0	0.0045	no_investigate	0
944	944	0	0.0079	no_investigate	0
945	945	0	0.0008	no_investigate	0
946	946	0	0.0424	no_investigate	0
947	947	0	0.0057	no_investigate	0
948	948	0	0.0002	no_investigate	0

949	949	0	0.0000	no_investigate	0
950	950	0	0.1560	investigate	500
951	951	0	0.0061	no_investigate	0
952	952	0	0.0080	no_investigate	0
953	953	0	0.0017	no_investigate	0
954	954	0	0.0001	no_investigate	0
955	955	1	0.0010	no_investigate	4500
956	956	1	0.6946	investigate	0
957	957	1	0.9887	investigate	0
958	958	0	0.0040	no_investigate	0
959	959	0	0.0040	no_investigate	0
960	960	0	0.0450	no_investigate	0
961	961	0	0.0376	no_investigate	0
962	962	0	0.0042	no_investigate	0
963	963	0	0.0000	no_investigate	0
964	964	0	0.0007	no_investigate	0
965	965	0	0.0048	no_investigate	0
966	966	0	0.0010	no_investigate	0
967	967	0	0.0042	no_investigate	0
968	968	0	0.0005	no_investigate	0
969	969	0	0.0004	no_investigate	0
970	970	0	0.0001	no_investigate	0
971	971	0	0.0007	no_investigate	0
972	972	1	0.0321	no_investigate	4500
973	973	1	0.4902	investigate	0
974	974	0	0.0003	no_investigate	0
975	975	0	0.0007	no_investigate	0
976	976	0	0.1551	investigate	500
977	977	0	0.0027	no_investigate	0
978	978	0	0.0035	no_investigate	0
979	979	0	0.0526	no_investigate	0
980	980	0	0.0003	no_investigate	0
981	981	0	0.0001	no_investigate	0
982	982	0	0.0280	no_investigate	0
983	983	0	0.0158	no_investigate	0
984	984	0	0.0094	no_investigate	0
985	985	0	0.0017	no_investigate	0
986	986	0	0.0018	no_investigate	0
987	987	0	0.0705	no_investigate	0
988	988	0	0.0269	no_investigate	0
989	989	0	0.0235	no_investigate	0
990	990	0	0.0823	no_investigate	0
991	991	0	0.0001	no_investigate	0
992	992	0	0.0392	no_investigate	0
993	993	0	0.0016	no_investigate	0
994	994	0	0.0007	no_investigate	0
995	995	0	0.0276	no_investigate	0
996	996	0	0.0007	no_investigate	0
997	997	0	0.0019	no_investigate	0
998	998	0	0.0014	no_investigate	0
999	999	0	0.0139	no_investigate	0

1000	1000	0	0.0003	no_investigate	0
1001	1001	0	0.0149	no_investigate	0
1002	1002	0	0.3902	investigate	500
1003	1003	0	0.0005	no_investigate	0
1004	1004	0	0.0010	no_investigate	0
1005	1005	0	0.0015	no_investigate	0
1006	1006	0	0.0003	no_investigate	0
1007	1007	0	0.0032	no_investigate	0
1008	1008	1	0.0532	no_investigate	4500
1009	1009	0	0.0067	no_investigate	0
1010	1010	0	0.0157	no_investigate	0
1011	1011	0	0.0031	no_investigate	0
1012	1012	0	0.0008	no_investigate	0
1013	1013	0	0.0007	no_investigate	0
1014	1014	0	0.0016	no_investigate	0
1015	1015	0	0.0007	no_investigate	0
1016	1016	0	0.0014	no_investigate	0
1017	1017	0	0.0003	no_investigate	0
1018	1018	0	0.0084	no_investigate	0
1019	1019	0	0.0131	no_investigate	0
1020	1020	0	0.0107	no_investigate	0
1021	1021	0	0.0032	no_investigate	0
1022	1022	0	0.0002	no_investigate	0
1023	1023	0	0.0001	no_investigate	0
1024	1024	0	0.0061	no_investigate	0
1025	1025	0	0.0037	no_investigate	0
1026	1026	0	0.0174	no_investigate	0
1027	1027	0	0.0495	no_investigate	0
1028	1028	0	0.0033	no_investigate	0
1029	1029	0	0.0016	no_investigate	0
1030	1030	0	0.0006	no_investigate	0
1031	1031	0	0.0008	no_investigate	0
1032	1032	0	0.0039	no_investigate	0
1033	1033	0	0.0444	no_investigate	0
1034	1034	1	0.0692	no_investigate	4500
1035	1035	0	0.0284	no_investigate	0
1036	1036	0	0.0005	no_investigate	0
1037	1037	0	0.0421	no_investigate	0
1038	1038	1	0.0735	no_investigate	4500
1039	1039	0	0.0002	no_investigate	0
1040	1040	0	0.0412	no_investigate	0
1041	1041	0	0.0084	no_investigate	0
1042	1042	0	0.0059	no_investigate	0
1043	1043	0	0.0401	no_investigate	0
1044	1044	0	0.0180	no_investigate	0
1045	1045	0	0.0882	no_investigate	0
1046	1046	0	0.0392	no_investigate	0
1047	1047	1	0.5634	investigate	0
1048	1048	0	0.0040	no_investigate	0
1049	1049	0	0.0001	no_investigate	0
1050	1050	1	0.2964	investigate	0

1051	1051	0	0.0042	no_investigate	0
1052	1052	1	0.0920	no_investigate	4500
1053	1053	0	0.0023	no_investigate	0
1054	1054	0	0.0065	no_investigate	0
1055	1055	0	0.0004	no_investigate	0
1056	1056	0	0.0586	no_investigate	0
1057	1057	0	0.0004	no_investigate	0
1058	1058	0	0.0044	no_investigate	0
1059	1059	0	0.0002	no_investigate	0
1060	1060	0	0.0095	no_investigate	0
1061	1061	1	0.0003	no_investigate	4500
1062	1062	0	0.0037	no_investigate	0
1063	1063	0	0.0135	no_investigate	0
1064	1064	1	0.2646	investigate	0
1065	1065	0	0.0039	no_investigate	0
1066	1066	0	0.0017	no_investigate	0
1067	1067	0	0.0003	no_investigate	0
1068	1068	0	0.0000	no_investigate	0
1069	1069	0	0.0002	no_investigate	0
1070	1070	0	0.0002	no_investigate	0
1071	1071	0	0.0024	no_investigate	0
1072	1072	0	0.0351	no_investigate	0
1073	1073	0	0.0002	no_investigate	0
1074	1074	0	0.0053	no_investigate	0
1075	1075	0	0.0000	no_investigate	0
1076	1076	0	0.0567	no_investigate	0
1077	1077	0	0.0050	no_investigate	0
1078	1078	0	0.0037	no_investigate	0
1079	1079	0	0.0015	no_investigate	0
1080	1080	0	0.0027	no_investigate	0
1081	1081	0	0.0141	no_investigate	0
1082	1082	0	0.0124	no_investigate	0
1083	1083	0	0.0126	no_investigate	0
1084	1084	0	0.0000	no_investigate	0
1085	1085	0	0.2910	investigate	500
1086	1086	0	0.0041	no_investigate	0
1087	1087	0	0.0306	no_investigate	0
1088	1088	0	0.0004	no_investigate	0
1089	1089	0	0.0025	no_investigate	0
1090	1090	0	0.0001	no_investigate	0
1091	1091	1	0.6963	investigate	0
1092	1092	0	0.0380	no_investigate	0
1093	1093	0	0.0100	no_investigate	0
1094	1094	0	0.0011	no_investigate	0
1095	1095	1	0.0516	no_investigate	4500
1096	1096	0	0.0034	no_investigate	0
1097	1097	0	0.0086	no_investigate	0
1098	1098	0	0.0075	no_investigate	0
1099	1099	1	0.1654	investigate	0
1100	1100	0	0.0326	no_investigate	0
1101	1101	0	0.0153	no_investigate	0

1102	1102	0	0.0000	no_investigate	0
1103	1103	0	0.1017	investigate	500
1104	1104	0	0.0095	no_investigate	0
1105	1105	0	0.0588	no_investigate	0
1106	1106	0	0.0259	no_investigate	0
1107	1107	0	0.0070	no_investigate	0
1108	1108	0	0.0014	no_investigate	0
1109	1109	0	0.0008	no_investigate	0
1110	1110	0	0.0079	no_investigate	0
1111	1111	0	0.0032	no_investigate	0
1112	1112	0	0.0041	no_investigate	0
1113	1113	0	0.0012	no_investigate	0
1114	1114	0	0.0066	no_investigate	0
1115	1115	0	0.0011	no_investigate	0
1116	1116	1	0.2762	investigate	0
1117	1117	0	0.0014	no_investigate	0
1118	1118	0	0.0002	no_investigate	0
1119	1119	0	0.0003	no_investigate	0
1120	1120	0	0.0641	no_investigate	0
1121	1121	0	0.0383	no_investigate	0
1122	1122	0	0.0010	no_investigate	0
1123	1123	0	0.0022	no_investigate	0
1124	1124	0	0.0001	no_investigate	0
1125	1125	0	0.0101	no_investigate	0
1126	1126	0	0.0275	no_investigate	0
1127	1127	0	0.0007	no_investigate	0
1128	1128	0	0.0002	no_investigate	0
1129	1129	0	0.0152	no_investigate	0
1130	1130	0	0.0590	no_investigate	0
1131	1131	0	0.0048	no_investigate	0
1132	1132	0	0.0049	no_investigate	0
1133	1133	0	0.0012	no_investigate	0
1134	1134	0	0.0001	no_investigate	0
1135	1135	0	0.0002	no_investigate	0
1136	1136	0	0.0001	no_investigate	0
1137	1137	0	0.0073	no_investigate	0
1138	1138	1	0.0230	no_investigate	4500
1139	1139	0	0.3887	investigate	500
1140	1140	1	0.0050	no_investigate	4500
1141	1141	0	0.0002	no_investigate	0
1142	1142	0	0.0042	no_investigate	0
1143	1143	0	0.0000	no_investigate	0
1144	1144	0	0.2113	investigate	500
1145	1145	0	0.0056	no_investigate	0
1146	1146	0	0.0480	no_investigate	0
1147	1147	0	0.0546	no_investigate	0
1148	1148	0	0.1806	investigate	500
1149	1149	0	0.1955	investigate	500
1150	1150	0	0.0013	no_investigate	0
1151	1151	0	0.0003	no_investigate	0
1152	1152	0	0.0180	no_investigate	0

1153	1153	0	0.2061	investigate	500
1154	1154	0	0.0179	no_investigate	0
1155	1155	0	0.1222	investigate	500
1156	1156	0	0.3872	investigate	500
1157	1157	0	0.0015	no_investigate	0
1158	1158	0	0.0006	no_investigate	0
1159	1159	0	0.0344	no_investigate	0
1160	1160	0	0.0204	no_investigate	0
1161	1161	0	0.0019	no_investigate	0
1162	1162	0	0.0061	no_investigate	0
1163	1163	0	0.0015	no_investigate	0
1164	1164	0	0.0061	no_investigate	0
1165	1165	0	0.0471	no_investigate	0
1166	1166	0	0.0012	no_investigate	0
1167	1167	1	0.7701	investigate	0
1168	1168	0	0.0050	no_investigate	0
1169	1169	0	0.2732	investigate	500
1170	1170	0	0.0003	no_investigate	0
1171	1171	0	0.0995	no_investigate	0
1172	1172	0	0.3541	investigate	500
1173	1173	0	0.0120	no_investigate	0
1174	1174	0	0.0233	no_investigate	0
1175	1175	0	0.0005	no_investigate	0
1176	1176	0	0.0111	no_investigate	0
1177	1177	0	0.0521	no_investigate	0
1178	1178	0	0.0022	no_investigate	0
1179	1179	0	0.0893	no_investigate	0
1180	1180	0	0.0070	no_investigate	0
1181	1181	0	0.0000	no_investigate	0
1182	1182	0	0.0003	no_investigate	0
1183	1183	0	0.0101	no_investigate	0
1184	1184	0	0.0045	no_investigate	0
1185	1185	0	0.0049	no_investigate	0
1186	1186	0	0.0009	no_investigate	0
1187	1187	0	0.0232	no_investigate	0
1188	1188	1	0.1921	investigate	0
1189	1189	0	0.0017	no_investigate	0
1190	1190	0	0.0009	no_investigate	0
1191	1191	0	0.1636	investigate	500
1192	1192	1	0.0804	no_investigate	4500
1193	1193	0	0.0001	no_investigate	0
1194	1194	0	0.0002	no_investigate	0
1195	1195	0	0.0063	no_investigate	0
1196	1196	0	0.0012	no_investigate	0
1197	1197	1	0.3293	investigate	0
1198	1198	0	0.0046	no_investigate	0
1199	1199	1	0.0182	no_investigate	4500
1200	1200	0	0.0485	no_investigate	0
1201	1201	0	0.0007	no_investigate	0
1202	1202	0	0.0207	no_investigate	0
1203	1203	0	0.0342	no_investigate	0

1204	1204	0	0.0708	no_investigate	0
1205	1205	0	0.0002	no_investigate	0
1206	1206	0	0.0007	no_investigate	0
1207	1207	0	0.0500	no_investigate	0
1208	1208	1	0.0017	no_investigate	4500
1209	1209	0	0.0007	no_investigate	0
1210	1210	0	0.2162	investigate	500
1211	1211	0	0.0319	no_investigate	0
1212	1212	0	0.0360	no_investigate	0
1213	1213	0	0.0038	no_investigate	0
1214	1214	0	0.0020	no_investigate	0
1215	1215	0	0.1428	investigate	500
1216	1216	0	0.0002	no_investigate	0
1217	1217	0	0.0001	no_investigate	0
1218	1218	0	0.0234	no_investigate	0
1219	1219	0	0.0022	no_investigate	0
1220	1220	0	0.0010	no_investigate	0
1221	1221	1	0.0524	no_investigate	4500
1222	1222	0	0.2766	investigate	500
1223	1223	0	0.0607	no_investigate	0
1224	1224	0	0.0044	no_investigate	0
1225	1225	1	0.2003	investigate	0
1226	1226	0	0.0416	no_investigate	0
1227	1227	0	0.0009	no_investigate	0
1228	1228	0	0.0099	no_investigate	0
1229	1229	1	0.0845	no_investigate	4500
1230	1230	0	0.0037	no_investigate	0
1231	1231	0	0.0075	no_investigate	0
1232	1232	0	0.0284	no_investigate	0
1233	1233	0	0.0029	no_investigate	0
1234	1234	0	0.1971	investigate	500
1235	1235	0	0.0620	no_investigate	0
1236	1236	0	0.0011	no_investigate	0
1237	1237	0	0.0025	no_investigate	0
1238	1238	0	0.0001	no_investigate	0
1239	1239	0	0.0001	no_investigate	0
1240	1240	0	0.0014	no_investigate	0
1241	1241	0	0.4112	investigate	500
1242	1242	0	0.0028	no_investigate	0
1243	1243	0	0.0093	no_investigate	0
1244	1244	0	0.5916	investigate	500
1245	1245	0	0.0073	no_investigate	0
1246	1246	0	0.0743	no_investigate	0
1247	1247	0	0.0087	no_investigate	0
1248	1248	0	0.0003	no_investigate	0
1249	1249	0	0.0001	no_investigate	0
1250	1250	0	0.0285	no_investigate	0
1251	1251	0	0.0875	no_investigate	0
1252	1252	0	0.0647	no_investigate	0
1253	1253	0	0.0059	no_investigate	0
1254	1254	0	0.0103	no_investigate	0

1255	1255	0	0.0002	no_investigate	0
1256	1256	0	0.0002	no_investigate	0
1257	1257	0	0.0004	no_investigate	0
1258	1258	0	0.0064	no_investigate	0
1259	1259	1	0.0659	no_investigate	4500
1260	1260	0	0.0003	no_investigate	0
1261	1261	1	0.0217	no_investigate	4500
1262	1262	0	0.0003	no_investigate	0
1263	1263	0	0.0183	no_investigate	0
1264	1264	0	0.0004	no_investigate	0
1265	1265	1	0.1867	investigate	0
1266	1266	0	0.0008	no_investigate	0
1267	1267	0	0.0415	no_investigate	0
1268	1268	0	0.0056	no_investigate	0
1269	1269	0	0.0281	no_investigate	0
1270	1270	0	0.3157	investigate	500
1271	1271	0	0.0003	no_investigate	0
1272	1272	0	0.0111	no_investigate	0
1273	1273	0	0.0386	no_investigate	0
1274	1274	0	0.0093	no_investigate	0
1275	1275	0	0.0002	no_investigate	0
1276	1276	0	0.0041	no_investigate	0
1277	1277	0	0.0260	no_investigate	0
1278	1278	0	0.0006	no_investigate	0
1279	1279	0	0.0000	no_investigate	0
1280	1280	0	0.0012	no_investigate	0
1281	1281	0	0.0166	no_investigate	0
1282	1282	0	0.0012	no_investigate	0
1283	1283	0	0.0007	no_investigate	0
1284	1284	0	0.1460	investigate	500
1285	1285	0	0.0126	no_investigate	0
1286	1286	0	0.0003	no_investigate	0
1287	1287	1	0.2232	investigate	0
1288	1288	0	0.0000	no_investigate	0
1289	1289	0	0.0233	no_investigate	0
1290	1290	0	0.6772	investigate	500
1291	1291	0	0.0005	no_investigate	0
1292	1292	0	0.0236	no_investigate	0
1293	1293	0	0.0006	no_investigate	0
1294	1294	0	0.0015	no_investigate	0
1295	1295	0	0.0002	no_investigate	0
1296	1296	0	0.0003	no_investigate	0
1297	1297	1	0.2588	investigate	0
1298	1298	0	0.0014	no_investigate	0
1299	1299	0	0.0002	no_investigate	0
1300	1300	0	0.0205	no_investigate	0
1301	1301	0	0.0002	no_investigate	0
1302	1302	0	0.0265	no_investigate	0
1303	1303	0	0.4399	investigate	500
1304	1304	0	0.1947	investigate	500
1305	1305	0	0.0014	no_investigate	0

1306	1306	1	0.0258	no_investigate	4500
1307	1307	0	0.0090	no_investigate	0
1308	1308	0	0.0021	no_investigate	0
1309	1309	0	0.0006	no_investigate	0
1310	1310	0	0.0278	no_investigate	0
1311	1311	0	0.0285	no_investigate	0
1312	1312	1	0.3894	investigate	0
1313	1313	0	0.0004	no_investigate	0
1314	1314	0	0.0074	no_investigate	0
1315	1315	0	0.0016	no_investigate	0
1316	1316	0	0.0002	no_investigate	0
1317	1317	0	0.0012	no_investigate	0
1318	1318	0	0.2287	investigate	500
1319	1319	0	0.0009	no_investigate	0
1320	1320	0	0.0092	no_investigate	0
1321	1321	1	0.0055	no_investigate	4500
1322	1322	0	0.0267	no_investigate	0
1323	1323	0	0.0021	no_investigate	0
1324	1324	0	0.1446	investigate	500
1325	1325	0	0.0141	no_investigate	0
1326	1326	0	0.0009	no_investigate	0
1327	1327	0	0.0129	no_investigate	0
1328	1328	0	0.0004	no_investigate	0
1329	1329	0	0.0170	no_investigate	0
1330	1330	0	0.0004	no_investigate	0
1331	1331	0	0.0040	no_investigate	0
1332	1332	1	0.0635	no_investigate	4500
1333	1333	0	0.0012	no_investigate	0
1334	1334	0	0.0102	no_investigate	0
1335	1335	0	0.0004	no_investigate	0
1336	1336	0	0.3196	investigate	500
1337	1337	0	0.0041	no_investigate	0
1338	1338	0	0.0030	no_investigate	0
1339	1339	0	0.0010	no_investigate	0
1340	1340	0	0.0091	no_investigate	0
1341	1341	1	0.7857	investigate	0
1342	1342	0	0.5250	investigate	500
1343	1343	0	0.0021	no_investigate	0
1344	1344	0	0.0003	no_investigate	0
1345	1345	0	0.0027	no_investigate	0
1346	1346	0	0.0003	no_investigate	0
1347	1347	0	0.0157	no_investigate	0
1348	1348	0	0.0002	no_investigate	0
1349	1349	0	0.0002	no_investigate	0
1350	1350	0	0.2971	investigate	500
1351	1351	0	0.0118	no_investigate	0
1352	1352	0	0.0360	no_investigate	0
1353	1353	0	0.0000	no_investigate	0
1354	1354	0	0.1312	investigate	500
1355	1355	0	0.0052	no_investigate	0
1356	1356	0	0.0003	no_investigate	0

1357	1357	0	0.0009	no_investigate	0
1358	1358	0	0.0002	no_investigate	0
1359	1359	0	0.0075	no_investigate	0
1360	1360	0	0.0005	no_investigate	0
1361	1361	0	0.0226	no_investigate	0
1362	1362	0	0.0000	no_investigate	0
1363	1363	0	0.0029	no_investigate	0
1364	1364	0	0.1586	investigate	500
1365	1365	0	0.0001	no_investigate	0
1366	1366	0	0.0005	no_investigate	0
1367	1367	0	0.0016	no_investigate	0
1368	1368	0	0.0001	no_investigate	0
1369	1369	0	0.0026	no_investigate	0
1370	1370	0	0.0018	no_investigate	0
1371	1371	0	0.0000	no_investigate	0
1372	1372	0	0.0005	no_investigate	0
1373	1373	0	0.0063	no_investigate	0
1374	1374	0	0.0412	no_investigate	0
1375	1375	0	0.0575	no_investigate	0
1376	1376	0	0.0630	no_investigate	0
1377	1377	0	0.0008	no_investigate	0
1378	1378	1	0.1453	investigate	0
1379	1379	0	0.2216	investigate	500
1380	1380	0	0.0163	no_investigate	0
1381	1381	0	0.0011	no_investigate	0
1382	1382	0	0.1162	investigate	500
1383	1383	0	0.0011	no_investigate	0
1384	1384	0	0.0006	no_investigate	0
1385	1385	0	0.0363	no_investigate	0
1386	1386	0	0.0004	no_investigate	0
1387	1387	0	0.0229	no_investigate	0
1388	1388	0	0.0886	no_investigate	0
1389	1389	0	0.0000	no_investigate	0
1390	1390	0	0.0011	no_investigate	0
1391	1391	0	0.0483	no_investigate	0
1392	1392	0	0.0500	no_investigate	0
1393	1393	0	0.0012	no_investigate	0
1394	1394	0	0.0003	no_investigate	0
1395	1395	1	0.7355	investigate	0
1396	1396	0	0.0008	no_investigate	0
1397	1397	0	0.0410	no_investigate	0
1398	1398	0	0.0002	no_investigate	0
1399	1399	1	0.0899	no_investigate	4500
1400	1400	0	0.0031	no_investigate	0
1401	1401	0	0.0876	no_investigate	0
1402	1402	0	0.0379	no_investigate	0
1403	1403	0	0.0001	no_investigate	0
1404	1404	0	0.0013	no_investigate	0
1405	1405	0	0.0356	no_investigate	0
1406	1406	0	0.0046	no_investigate	0
1407	1407	0	0.0006	no_investigate	0

1408	1408	0	0.0122	no_investigate	0
1409	1409	0	0.0030	no_investigate	0
1410	1410	0	0.0012	no_investigate	0
1411	1411	0	0.0004	no_investigate	0
1412	1412	0	0.0001	no_investigate	0
1413	1413	0	0.0015	no_investigate	0
1414	1414	0	0.0003	no_investigate	0
1415	1415	0	0.0007	no_investigate	0
1416	1416	0	0.0004	no_investigate	0
1417	1417	0	0.0465	no_investigate	0
1418	1418	0	0.0096	no_investigate	0
1419	1419	0	0.1152	investigate	500
1420	1420	1	0.0653	no_investigate	4500
1421	1421	0	0.0128	no_investigate	0
1422	1422	0	0.0001	no_investigate	0
1423	1423	0	0.0001	no_investigate	0
1424	1424	0	0.0087	no_investigate	0
1425	1425	0	0.0004	no_investigate	0
1426	1426	1	0.0005	no_investigate	4500
1427	1427	0	0.1088	investigate	500
1428	1428	0	0.0013	no_investigate	0
1429	1429	0	0.0002	no_investigate	0
1430	1430	0	0.0051	no_investigate	0
1431	1431	0	0.0087	no_investigate	0
1432	1432	0	0.0010	no_investigate	0
1433	1433	0	0.0075	no_investigate	0
1434	1434	0	0.0030	no_investigate	0
1435	1435	0	0.0032	no_investigate	0
1436	1436	0	0.0002	no_investigate	0
1437	1437	0	0.0002	no_investigate	0
1438	1438	0	0.0468	no_investigate	0
1439	1439	0	0.0004	no_investigate	0
1440	1440	0	0.0016	no_investigate	0
1441	1441	0	0.0094	no_investigate	0
1442	1442	0	0.0002	no_investigate	0
1443	1443	0	0.0000	no_investigate	0
1444	1444	0	0.0571	no_investigate	0
1445	1445	1	0.2202	investigate	0
1446	1446	0	0.0020	no_investigate	0
1447	1447	0	0.0068	no_investigate	0
1448	1448	1	0.2114	investigate	0
1449	1449	0	0.2023	investigate	500
1450	1450	0	0.0015	no_investigate	0
1451	1451	0	0.0032	no_investigate	0
1452	1452	0	0.0033	no_investigate	0
1453	1453	0	0.0214	no_investigate	0
1454	1454	0	0.1171	investigate	500
1455	1455	1	0.5685	investigate	0
1456	1456	0	0.0002	no_investigate	0
1457	1457	0	0.0012	no_investigate	0
1458	1458	0	0.0005	no_investigate	0

1459	1459	0	0.0261	no_investigate	0
1460	1460	0	0.0102	no_investigate	0
1461	1461	0	0.0003	no_investigate	0
1462	1462	0	0.0002	no_investigate	0
1463	1463	1	0.0787	no_investigate	4500
1464	1464	0	0.0000	no_investigate	0
1465	1465	0	0.0016	no_investigate	0
1466	1466	0	0.0221	no_investigate	0
1467	1467	0	0.0001	no_investigate	0
1468	1468	0	0.0025	no_investigate	0
1469	1469	0	0.0011	no_investigate	0
1470	1470	0	0.0771	no_investigate	0
1471	1471	0	0.0063	no_investigate	0
1472	1472	0	0.1098	investigate	500
1473	1473	0	0.0077	no_investigate	0
1474	1474	0	0.0014	no_investigate	0
1475	1475	0	0.1801	investigate	500
1476	1476	0	0.0208	no_investigate	0
1477	1477	1	0.0420	no_investigate	4500
1478	1478	0	0.0038	no_investigate	0
1479	1479	0	0.0021	no_investigate	0
1480	1480	0	0.0787	no_investigate	0
1481	1481	0	0.0033	no_investigate	0
1482	1482	0	0.0139	no_investigate	0
1483	1483	0	0.0023	no_investigate	0
1484	1484	0	0.0002	no_investigate	0
1485	1485	0	0.0310	no_investigate	0
1486	1486	0	0.0987	no_investigate	0
1487	1487	0	0.0012	no_investigate	0
1488	1488	0	0.0016	no_investigate	0
1489	1489	0	0.0197	no_investigate	0
1490	1490	1	0.0952	no_investigate	4500
1491	1491	0	0.0487	no_investigate	0
1492	1492	0	0.0004	no_investigate	0
1493	1493	1	0.0306	no_investigate	4500
1494	1494	0	0.0010	no_investigate	0
1495	1495	0	0.0031	no_investigate	0
1496	1496	0	0.0464	no_investigate	0
1497	1497	0	0.0014	no_investigate	0
1498	1498	0	0.0159	no_investigate	0
1499	1499	0	0.0001	no_investigate	0
1500	1500	0	0.0035	no_investigate	0
1501	1501	0	0.0774	no_investigate	0
1502	1502	0	0.0001	no_investigate	0
1503	1503	0	0.0056	no_investigate	0
1504	1504	0	0.0012	no_investigate	0
1505	1505	0	0.0013	no_investigate	0
1506	1506	0	0.0020	no_investigate	0
1507	1507	0	0.0027	no_investigate	0
1508	1508	0	0.0184	no_investigate	0
1509	1509	0	0.0001	no_investigate	0

1510	1510	0	0.0028	no_investigate	0
1511	1511	0	0.0009	no_investigate	0
1512	1512	0	0.0013	no_investigate	0
1513	1513	0	0.0087	no_investigate	0
1514	1514	0	0.0019	no_investigate	0
1515	1515	0	0.0114	no_investigate	0
1516	1516	1	0.0049	no_investigate	4500
1517	1517	0	0.0258	no_investigate	0
1518	1518	0	0.0003	no_investigate	0
1519	1519	0	0.0030	no_investigate	0
1520	1520	1	0.6100	investigate	0
1521	1521	0	0.0042	no_investigate	0
1522	1522	0	0.0007	no_investigate	0
1523	1523	0	0.0000	no_investigate	0
1524	1524	0	0.0013	no_investigate	0
1525	1525	0	0.0001	no_investigate	0
1526	1526	0	0.0139	no_investigate	0
1527	1527	0	0.1039	investigate	500
1528	1528	0	0.0173	no_investigate	0
1529	1529	0	0.0005	no_investigate	0
1530	1530	0	0.0000	no_investigate	0
1531	1531	0	0.0000	no_investigate	0
1532	1532	0	0.0028	no_investigate	0
1533	1533	0	0.0045	no_investigate	0
1534	1534	0	0.0263	no_investigate	0
1535	1535	0	0.0864	no_investigate	0
1536	1536	0	0.0759	no_investigate	0
1537	1537	0	0.0067	no_investigate	0
1538	1538	0	0.0043	no_investigate	0
1539	1539	0	0.0008	no_investigate	0
1540	1540	0	0.0114	no_investigate	0
1541	1541	0	0.0001	no_investigate	0
1542	1542	0	0.0125	no_investigate	0
1543	1543	0	0.0038	no_investigate	0
1544	1544	0	0.0019	no_investigate	0
1545	1545	0	0.0064	no_investigate	0
1546	1546	0	0.0485	no_investigate	0
1547	1547	0	0.0030	no_investigate	0
1548	1548	0	0.0269	no_investigate	0
1549	1549	0	0.0446	no_investigate	0
1550	1550	0	0.0013	no_investigate	0
1551	1551	0	0.0988	no_investigate	0
1552	1552	0	0.0407	no_investigate	0
1553	1553	0	0.0007	no_investigate	0
1554	1554	0	0.0053	no_investigate	0
1555	1555	0	0.0010	no_investigate	0
1556	1556	1	0.2455	investigate	0
1557	1557	0	0.0126	no_investigate	0
1558	1558	0	0.0020	no_investigate	0
1559	1559	0	0.0310	no_investigate	0
1560	1560	0	0.0009	no_investigate	0

1561	1561	0	0.0001	no_investigate	0
1562	1562	0	0.0001	no_investigate	0
1563	1563	0	0.0151	no_investigate	0
1564	1564	0	0.0009	no_investigate	0
1565	1565	0	0.0050	no_investigate	0
1566	1566	0	0.0055	no_investigate	0
1567	1567	0	0.0494	no_investigate	0
1568	1568	0	0.0480	no_investigate	0
1569	1569	0	0.0067	no_investigate	0
1570	1570	0	0.0225	no_investigate	0
1571	1571	0	0.0039	no_investigate	0
1572	1572	0	0.0036	no_investigate	0
1573	1573	0	0.0027	no_investigate	0
1574	1574	0	0.0408	no_investigate	0
1575	1575	0	0.0062	no_investigate	0
1576	1576	0	0.0440	no_investigate	0
1577	1577	0	0.0075	no_investigate	0
1578	1578	0	0.0004	no_investigate	0
1579	1579	0	0.0001	no_investigate	0
1580	1580	0	0.0003	no_investigate	0
1581	1581	0	0.0763	no_investigate	0
1582	1582	0	0.0482	no_investigate	0
1583	1583	0	0.0400	no_investigate	0
1584	1584	0	0.0004	no_investigate	0
1585	1585	0	0.0012	no_investigate	0
1586	1586	0	0.0053	no_investigate	0
1587	1587	0	0.0004	no_investigate	0
1588	1588	0	0.0540	no_investigate	0
1589	1589	0	0.0005	no_investigate	0
1590	1590	0	0.0088	no_investigate	0
1591	1591	0	0.0003	no_investigate	0
1592	1592	0	0.0049	no_investigate	0
1593	1593	0	0.0025	no_investigate	0
1594	1594	0	0.0030	no_investigate	0
1595	1595	0	0.0059	no_investigate	0
1596	1596	0	0.0003	no_investigate	0
1597	1597	0	0.0158	no_investigate	0
1598	1598	0	0.0125	no_investigate	0
1599	1599	0	0.0000	no_investigate	0
1600	1600	0	0.0010	no_investigate	0
1601	1601	0	0.0026	no_investigate	0
1602	1602	0	0.0078	no_investigate	0
1603	1603	0	0.0004	no_investigate	0
1604	1604	0	0.0041	no_investigate	0
1605	1605	0	0.0037	no_investigate	0
1606	1606	0	0.0169	no_investigate	0
1607	1607	0	0.0496	no_investigate	0
1608	1608	0	0.0052	no_investigate	0
1609	1609	0	0.0055	no_investigate	0
1610	1610	0	0.0538	no_investigate	0
1611	1611	0	0.0436	no_investigate	0

1612	1612	0	0.0113	no_investigate	0
1613	1613	0	0.0193	no_investigate	0
1614	1614	0	0.0127	no_investigate	0
1615	1615	0	0.0024	no_investigate	0
1616	1616	0	0.0003	no_investigate	0
1617	1617	0	0.1052	investigate	500
1618	1618	0	0.0283	no_investigate	0
1619	1619	0	0.0004	no_investigate	0
1620	1620	0	0.0070	no_investigate	0
1621	1621	0	0.0015	no_investigate	0
1622	1622	0	0.0031	no_investigate	0
1623	1623	0	0.0664	no_investigate	0
1624	1624	0	0.0804	no_investigate	0
1625	1625	0	0.0235	no_investigate	0
1626	1626	0	0.0172	no_investigate	0
1627	1627	0	0.0031	no_investigate	0
1628	1628	0	0.0010	no_investigate	0
1629	1629	0	0.0839	no_investigate	0
1630	1630	0	0.0085	no_investigate	0
1631	1631	0	0.0017	no_investigate	0
1632	1632	0	0.0150	no_investigate	0
1633	1633	0	0.0108	no_investigate	0
1634	1634	0	0.0064	no_investigate	0
1635	1635	0	0.0158	no_investigate	0
1636	1636	0	0.2626	investigate	500
1637	1637	0	0.0966	no_investigate	0
1638	1638	0	0.0123	no_investigate	0
1639	1639	0	0.3636	investigate	500
1640	1640	0	0.0526	no_investigate	0
1641	1641	0	0.0002	no_investigate	0
1642	1642	0	0.0183	no_investigate	0
1643	1643	0	0.0010	no_investigate	0
1644	1644	0	0.0005	no_investigate	0
1645	1645	0	0.0024	no_investigate	0
1646	1646	0	0.0003	no_investigate	0
1647	1647	0	0.0349	no_investigate	0
1648	1648	1	0.0072	no_investigate	4500
1649	1649	0	0.0824	no_investigate	0
1650	1650	0	0.0035	no_investigate	0
1651	1651	0	0.0369	no_investigate	0
1652	1652	0	0.0028	no_investigate	0
1653	1653	0	0.0002	no_investigate	0
1654	1654	0	0.0657	no_investigate	0
1655	1655	0	0.0023	no_investigate	0
1656	1656	0	0.0034	no_investigate	0
1657	1657	0	0.1815	investigate	500
1658	1658	0	0.0203	no_investigate	0
1659	1659	0	0.0004	no_investigate	0
1660	1660	0	0.0027	no_investigate	0
1661	1661	0	0.0002	no_investigate	0
1662	1662	0	0.0052	no_investigate	0

1663	1663	0	0.0012	no_investigate	0
1664	1664	0	0.1100	investigate	500
1665	1665	0	0.0213	no_investigate	0
1666	1666	0	0.0033	no_investigate	0
1667	1667	0	0.0037	no_investigate	0
1668	1668	0	0.0009	no_investigate	0
1669	1669	0	0.4818	investigate	500
1670	1670	0	0.0197	no_investigate	0
1671	1671	0	0.0608	no_investigate	0
1672	1672	0	0.0062	no_investigate	0
1673	1673	0	0.0591	no_investigate	0
1674	1674	0	0.0002	no_investigate	0
1675	1675	0	0.0029	no_investigate	0
1676	1676	0	0.4798	investigate	500
1677	1677	0	0.0005	no_investigate	0
1678	1678	0	0.0109	no_investigate	0
1679	1679	0	0.1136	investigate	500
1680	1680	1	0.0253	no_investigate	4500
1681	1681	0	0.0191	no_investigate	0
1682	1682	0	0.0031	no_investigate	0
1683	1683	0	0.0004	no_investigate	0
1684	1684	0	0.0037	no_investigate	0
1685	1685	0	0.0026	no_investigate	0
1686	1686	0	0.0002	no_investigate	0
1687	1687	0	0.0036	no_investigate	0
1688	1688	0	0.0018	no_investigate	0
1689	1689	0	0.0041	no_investigate	0
1690	1690	0	0.0091	no_investigate	0
1691	1691	0	0.0082	no_investigate	0
1692	1692	0	0.0008	no_investigate	0
1693	1693	0	0.0106	no_investigate	0
1694	1694	0	0.0007	no_investigate	0
1695	1695	0	0.0264	no_investigate	0
1696	1696	0	0.0120	no_investigate	0
1697	1697	0	0.0169	no_investigate	0
1698	1698	0	0.0107	no_investigate	0
1699	1699	0	0.0024	no_investigate	0
1700	1700	0	0.0449	no_investigate	0
1701	1701	0	0.0003	no_investigate	0
1702	1702	0	0.0009	no_investigate	0
1703	1703	0	0.0012	no_investigate	0
1704	1704	0	0.0577	no_investigate	0
1705	1705	0	0.0185	no_investigate	0
1706	1706	0	0.0074	no_investigate	0
1707	1707	0	0.0002	no_investigate	0
1708	1708	0	0.0243	no_investigate	0
1709	1709	0	0.0003	no_investigate	0
1710	1710	0	0.0536	no_investigate	0
1711	1711	0	0.1706	investigate	500
1712	1712	0	0.0030	no_investigate	0
1713	1713	0	0.4911	investigate	500

1714	1714	0	0.0027	no_investigate	0
1715	1715	0	0.0010	no_investigate	0
1716	1716	0	0.0101	no_investigate	0
1717	1717	0	0.0221	no_investigate	0
1718	1718	1	0.1238	investigate	0
1719	1719	0	0.0003	no_investigate	0
1720	1720	0	0.0001	no_investigate	0
1721	1721	0	0.0015	no_investigate	0
1722	1722	0	0.0004	no_investigate	0
1723	1723	0	0.0001	no_investigate	0
1724	1724	0	0.0039	no_investigate	0
1725	1725	0	0.0079	no_investigate	0
1726	1726	0	0.0005	no_investigate	0
1727	1727	0	0.0403	no_investigate	0
1728	1728	0	0.0128	no_investigate	0
1729	1729	1	0.0429	no_investigate	4500
1730	1730	0	0.0043	no_investigate	0
1731	1731	0	0.0355	no_investigate	0
1732	1732	0	0.0072	no_investigate	0
1733	1733	0	0.0100	no_investigate	0
1734	1734	0	0.0273	no_investigate	0
1735	1735	0	0.0025	no_investigate	0
1736	1736	0	0.0536	no_investigate	0
1737	1737	0	0.0006	no_investigate	0
1738	1738	0	0.0893	no_investigate	0
1739	1739	0	0.0050	no_investigate	0
1740	1740	0	0.0051	no_investigate	0
1741	1741	0	0.0001	no_investigate	0
1742	1742	0	0.0115	no_investigate	0
1743	1743	1	0.7448	investigate	0
1744	1744	0	0.0270	no_investigate	0
1745	1745	0	0.0894	no_investigate	0
1746	1746	0	0.0037	no_investigate	0
1747	1747	0	0.0540	no_investigate	0
1748	1748	0	0.0026	no_investigate	0
1749	1749	0	0.0215	no_investigate	0
1750	1750	0	0.0029	no_investigate	0
1751	1751	0	0.0006	no_investigate	0
1752	1752	0	0.0002	no_investigate	0
1753	1753	0	0.0001	no_investigate	0
1754	1754	0	0.0377	no_investigate	0
1755	1755	0	0.0269	no_investigate	0
1756	1756	0	0.0000	no_investigate	0
1757	1757	0	0.0009	no_investigate	0
1758	1758	0	0.0034	no_investigate	0
1759	1759	0	0.0072	no_investigate	0
1760	1760	0	0.0002	no_investigate	0
1761	1761	0	0.0039	no_investigate	0
1762	1762	0	0.0002	no_investigate	0
1763	1763	0	0.0012	no_investigate	0
1764	1764	0	0.0002	no_investigate	0

1765	1765	0	0.0004	no_investigate	0
1766	1766	0	0.0037	no_investigate	0
1767	1767	0	0.0019	no_investigate	0
1768	1768	0	0.0072	no_investigate	0
1769	1769	0	0.0031	no_investigate	0
1770	1770	0	0.0062	no_investigate	0
1771	1771	1	0.8910	investigate	0
1772	1772	0	0.0019	no_investigate	0
1773	1773	0	0.0007	no_investigate	0
1774	1774	0	0.0199	no_investigate	0
1775	1775	0	0.1673	investigate	500
1776	1776	0	0.0034	no_investigate	0
1777	1777	0	0.0138	no_investigate	0
1778	1778	0	0.0141	no_investigate	0
1779	1779	0	0.0351	no_investigate	0
1780	1780	0	0.0017	no_investigate	0
1781	1781	0	0.0001	no_investigate	0
1782	1782	0	0.1315	investigate	500
1783	1783	0	0.0000	no_investigate	0
1784	1784	0	0.0000	no_investigate	0
1785	1785	0	0.4508	investigate	500
1786	1786	0	0.0040	no_investigate	0
1787	1787	0	0.0064	no_investigate	0
1788	1788	0	0.0056	no_investigate	0
1789	1789	0	0.1066	investigate	500
1790	1790	0	0.0002	no_investigate	0
1791	1791	0	0.0021	no_investigate	0
1792	1792	0	0.0107	no_investigate	0
1793	1793	0	0.0090	no_investigate	0
1794	1794	0	0.1336	investigate	500
1795	1795	0	0.0039	no_investigate	0
1796	1796	0	0.0014	no_investigate	0
1797	1797	0	0.0170	no_investigate	0
1798	1798	0	0.2565	investigate	500
1799	1799	0	0.0001	no_investigate	0
1800	1800	0	0.0012	no_investigate	0
1801	1801	0	0.0084	no_investigate	0
1802	1802	0	0.0025	no_investigate	0
1803	1803	1	0.1344	investigate	0
1804	1804	0	0.0048	no_investigate	0
1805	1805	0	0.0006	no_investigate	0
1806	1806	0	0.2842	investigate	500
1807	1807	0	0.0237	no_investigate	0
1808	1808	1	0.5768	investigate	0
1809	1809	0	0.0081	no_investigate	0
1810	1810	0	0.0217	no_investigate	0
1811	1811	1	0.2478	investigate	0
1812	1812	0	0.0028	no_investigate	0
1813	1813	0	0.0011	no_investigate	0
1814	1814	0	0.0000	no_investigate	0
1815	1815	0	0.0031	no_investigate	0

1816	1816	0	0.0268	no_investigate	0
1817	1817	0	0.0029	no_investigate	0
1818	1818	0	0.0831	no_investigate	0
1819	1819	0	0.5513	investigate	500
1820	1820	0	0.0010	no_investigate	0
1821	1821	0	0.0004	no_investigate	0
1822	1822	0	0.0086	no_investigate	0
1823	1823	0	0.0017	no_investigate	0
1824	1824	0	0.0003	no_investigate	0
1825	1825	0	0.0072	no_investigate	0
1826	1826	0	0.0034	no_investigate	0
1827	1827	0	0.0002	no_investigate	0
1828	1828	0	0.0570	no_investigate	0
1829	1829	1	0.7623	investigate	0
1830	1830	0	0.0128	no_investigate	0
1831	1831	0	0.0006	no_investigate	0
1832	1832	0	0.0003	no_investigate	0
1833	1833	0	0.0231	no_investigate	0
1834	1834	0	0.0015	no_investigate	0
1835	1835	0	0.0385	no_investigate	0
1836	1836	1	0.3732	investigate	0
1837	1837	0	0.0089	no_investigate	0
1838	1838	0	0.0123	no_investigate	0
1839	1839	1	0.2404	investigate	0
1840	1840	0	0.0003	no_investigate	0
1841	1841	0	0.0001	no_investigate	0
1842	1842	0	0.3188	investigate	500
1843	1843	0	0.0000	no_investigate	0
1844	1844	0	0.0390	no_investigate	0
1845	1845	0	0.0341	no_investigate	0
1846	1846	0	0.2853	investigate	500
1847	1847	1	0.0036	no_investigate	4500
1848	1848	0	0.0163	no_investigate	0
1849	1849	0	0.0000	no_investigate	0
1850	1850	0	0.1429	investigate	500
1851	1851	0	0.0019	no_investigate	0
1852	1852	0	0.0051	no_investigate	0
1853	1853	0	0.5738	investigate	500
1854	1854	0	0.0033	no_investigate	0
1855	1855	0	0.0045	no_investigate	0
1856	1856	0	0.0922	no_investigate	0
1857	1857	0	0.0013	no_investigate	0
1858	1858	0	0.0028	no_investigate	0
1859	1859	0	0.0086	no_investigate	0
1860	1860	0	0.0035	no_investigate	0
1861	1861	0	0.0437	no_investigate	0
1862	1862	0	0.0055	no_investigate	0
1863	1863	0	0.0020	no_investigate	0
1864	1864	0	0.1041	investigate	500
1865	1865	0	0.1022	investigate	500
1866	1866	0	0.0003	no_investigate	0

1867	1867	0	0.1226	investigate	500
1868	1868	0	0.0211	no_investigate	0
1869	1869	1	0.2547	investigate	0
1870	1870	0	0.0005	no_investigate	0
1871	1871	0	0.0002	no_investigate	0
1872	1872	0	0.0316	no_investigate	0
1873	1873	0	0.0300	no_investigate	0
1874	1874	0	0.1288	investigate	500
1875	1875	0	0.0009	no_investigate	0
1876	1876	0	0.0124	no_investigate	0
1877	1877	0	0.2003	investigate	500
1878	1878	0	0.0012	no_investigate	0
1879	1879	0	0.0003	no_investigate	0
1880	1880	0	0.0531	no_investigate	0
1881	1881	0	0.2069	investigate	500
1882	1882	0	0.0014	no_investigate	0
1883	1883	0	0.0010	no_investigate	0
1884	1884	0	0.0090	no_investigate	0
1885	1885	0	0.0015	no_investigate	0
1886	1886	0	0.0210	no_investigate	0
1887	1887	0	0.1052	investigate	500
1888	1888	0	0.1050	investigate	500
1889	1889	0	0.0001	no_investigate	0
1890	1890	0	0.0001	no_investigate	0
1891	1891	0	0.0069	no_investigate	0
1892	1892	0	0.0048	no_investigate	0
1893	1893	0	0.0122	no_investigate	0
1894	1894	0	0.2720	investigate	500
1895	1895	0	0.0017	no_investigate	0
1896	1896	0	0.0019	no_investigate	0
1897	1897	0	0.0001	no_investigate	0
1898	1898	0	0.0004	no_investigate	0
1899	1899	0	0.0055	no_investigate	0
1900	1900	0	0.0079	no_investigate	0
1901	1901	0	0.0011	no_investigate	0
1902	1902	0	0.0004	no_investigate	0
1903	1903	0	0.0071	no_investigate	0
1904	1904	0	0.3094	investigate	500
1905	1905	1	0.0008	no_investigate	4500
1906	1906	0	0.0386	no_investigate	0
1907	1907	0	0.1810	investigate	500
1908	1908	0	0.0059	no_investigate	0
1909	1909	1	0.4810	investigate	0
1910	1910	1	0.1884	investigate	0
1911	1911	0	0.0000	no_investigate	0
1912	1912	0	0.0351	no_investigate	0
1913	1913	0	0.4399	investigate	500
1914	1914	0	0.0242	no_investigate	0
1915	1915	0	0.0015	no_investigate	0
1916	1916	0	0.0404	no_investigate	0
1917	1917	0	0.0037	no_investigate	0

1918	1918	0	0.0002	no_investigate	0
1919	1919	0	0.0024	no_investigate	0
1920	1920	0	0.0089	no_investigate	0
1921	1921	0	0.0008	no_investigate	0
1922	1922	0	0.0662	no_investigate	0
1923	1923	0	0.0229	no_investigate	0
1924	1924	1	0.1452	investigate	0
1925	1925	0	0.0004	no_investigate	0
1926	1926	0	0.0009	no_investigate	0
1927	1927	0	0.0000	no_investigate	0
1928	1928	0	0.0417	no_investigate	0
1929	1929	0	0.0085	no_investigate	0
1930	1930	0	0.0002	no_investigate	0
1931	1931	0	0.1410	investigate	500
1932	1932	0	0.0002	no_investigate	0
1933	1933	0	0.0796	no_investigate	0
1934	1934	0	0.0120	no_investigate	0
1935	1935	0	0.0407	no_investigate	0
1936	1936	1	0.0727	no_investigate	4500
1937	1937	0	0.0187	no_investigate	0
1938	1938	0	0.0028	no_investigate	0
1939	1939	0	0.0222	no_investigate	0
1940	1940	1	0.0024	no_investigate	4500
1941	1941	0	0.0421	no_investigate	0
1942	1942	0	0.0022	no_investigate	0
1943	1943	0	0.0082	no_investigate	0
1944	1944	0	0.0014	no_investigate	0
1945	1945	0	0.0063	no_investigate	0
1946	1946	0	0.0013	no_investigate	0
1947	1947	0	0.0910	no_investigate	0
1948	1948	0	0.0005	no_investigate	0
1949	1949	0	0.0107	no_investigate	0
1950	1950	0	0.0016	no_investigate	0
1951	1951	0	0.0193	no_investigate	0
1952	1952	0	0.0008	no_investigate	0
1953	1953	0	0.0220	no_investigate	0
1954	1954	0	0.0008	no_investigate	0
1955	1955	0	0.0001	no_investigate	0
1956	1956	0	0.0010	no_investigate	0
1957	1957	0	0.0038	no_investigate	0
1958	1958	0	0.0148	no_investigate	0
1959	1959	0	0.0027	no_investigate	0
1960	1960	0	0.0333	no_investigate	0
1961	1961	0	0.0025	no_investigate	0
1962	1962	0	0.0033	no_investigate	0
1963	1963	0	0.0219	no_investigate	0
1964	1964	0	0.0264	no_investigate	0
1965	1965	0	0.0132	no_investigate	0
1966	1966	0	0.0000	no_investigate	0
1967	1967	0	0.0464	no_investigate	0
1968	1968	0	0.0021	no_investigate	0

1969	1969	0	0.0033	no_investigate	0
1970	1970	0	0.0126	no_investigate	0
1971	1971	0	0.0173	no_investigate	0
1972	1972	0	0.0040	no_investigate	0
1973	1973	0	0.0037	no_investigate	0
1974	1974	0	0.0031	no_investigate	0
1975	1975	0	0.0544	no_investigate	0
1976	1976	0	0.0513	no_investigate	0
1977	1977	0	0.1008	investigate	500
1978	1978	0	0.0074	no_investigate	0
1979	1979	0	0.2819	investigate	500
1980	1980	0	0.0003	no_investigate	0
1981	1981	0	0.0006	no_investigate	0
1982	1982	0	0.1046	investigate	500
1983	1983	0	0.0008	no_investigate	0
1984	1984	0	0.1187	investigate	500
1985	1985	0	0.0037	no_investigate	0
1986	1986	0	0.0532	no_investigate	0
1987	1987	0	0.0005	no_investigate	0
1988	1988	0	0.0049	no_investigate	0
1989	1989	0	0.0561	no_investigate	0
1990	1990	0	0.0018	no_investigate	0
1991	1991	0	0.0005	no_investigate	0
1992	1992	0	0.0050	no_investigate	0
1993	1993	1	0.2489	investigate	0
1994	1994	0	0.0002	no_investigate	0
1995	1995	0	0.0738	no_investigate	0
1996	1996	0	0.0093	no_investigate	0
1997	1997	0	0.0610	no_investigate	0
1998	1998	0	0.0010	no_investigate	0
1999	1999	0	0.0212	no_investigate	0
2000	2000	0	0.0438	no_investigate	0

```
# Total cost
sum(df_cost$cost)
```

```
[1] 403500
```

Observation: Using a 10% threshold → total cost = \$403,500 (for 2,000 claims)

Step 3: Using the cost function to try out different thresholds other than just 10%

```
thresholds <- seq(0, 1, by = 0.01)
```

```

results <- data.frame(threshold = thresholds, total_cost = NA)

for (i in seq_along(thresholds)) {

  t <- thresholds[i]

  df_temp <- df %>%
    mutate(
      decision = ifelse(pred_prob > t, "investigate", "no_investigate"),

      cost = case_when(
        decision == "investigate" & true_label == 0 ~ 500,
        decision == "investigate" & true_label == 1 ~ 0,
        decision == "no_investigate" & true_label == 1 ~ 4500,
        decision == "no_investigate" & true_label == 0 ~ 0
      )
    )

  results$total_cost[i] <- sum(df_temp$cost)
}
results[which.min(results$total_cost), ]

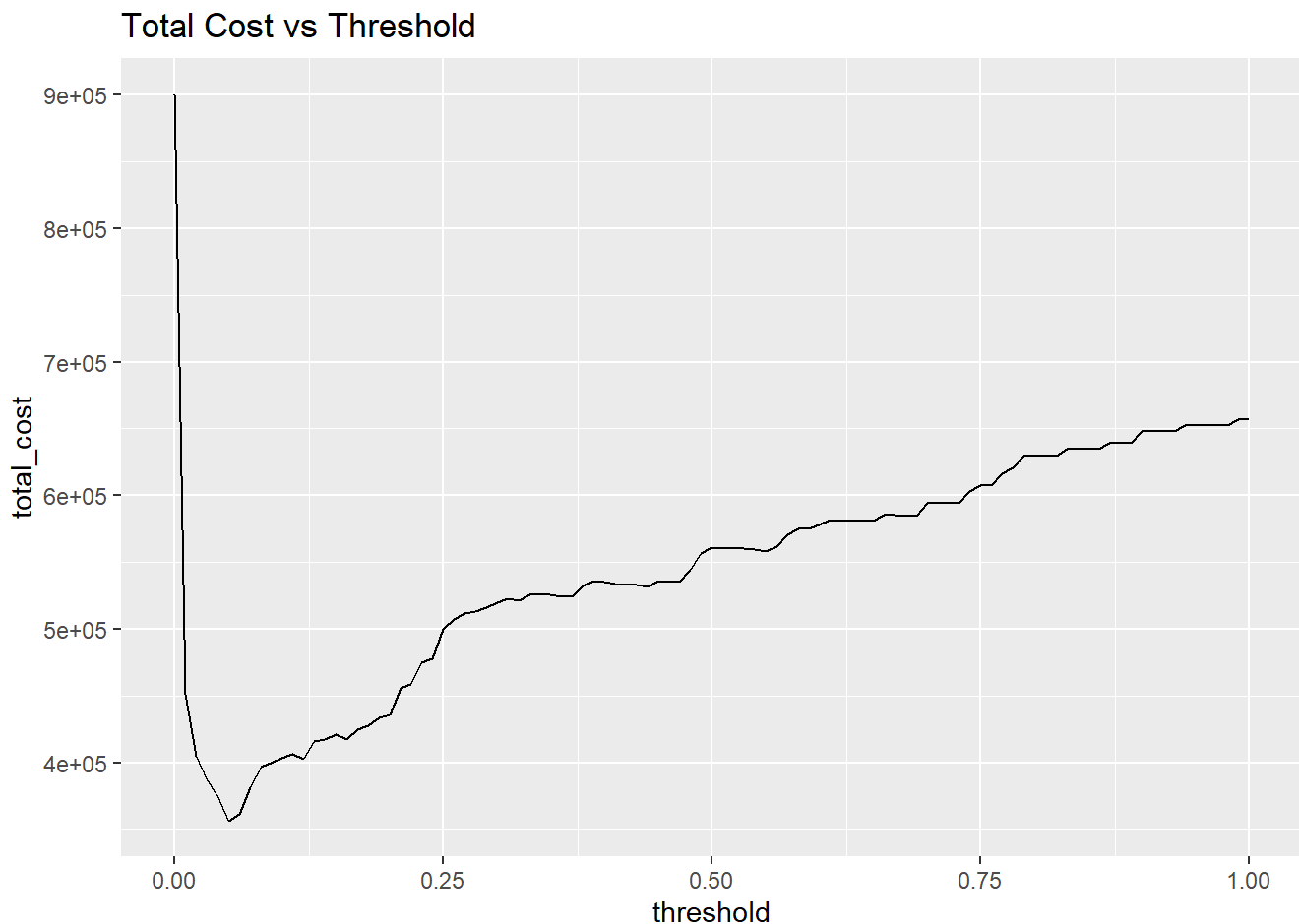
```

	threshold	total_cost
6	0.05	356500

```

ggplot(results, aes(x = threshold, y = total_cost)) +
  geom_line() +
  labs(title = "Total Cost vs Threshold")

```



Observation: While the theoretical threshold is 10%, analysis of actual claim outcomes shows that a lower threshold of 5% minimizes total cost. This suggests that the current workflow is too conservative and may be missing costly fraudulent claims.

```
ggplot(results, aes(x = threshold, y = total_cost)) +
  geom_line(size = 1) +

  geom_vline(xintercept = 0.05, linetype = "dashed", color = "blue") +
  geom_vline(xintercept = 0.10, linetype = "dashed", color = "red") +

  annotate("text", x = 0.05, y = 370000,
    label = "Optimal (~5%)", color = "blue") +

  annotate("text", x = 0.10, y = 420000,
    label = "Current (10%)", color = "red") +

  labs(
    title = "Total Cost vs Threshold",
    x = "Threshold",
    y = "Total Cost"
```

```
) +  
theme_minimal()
```

